MATHEMATICS GRADE 10: PROBABILITY I

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 4.2 (9 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 4.0: Data Handling and Probability
Sub-Strand	Sub-Strand 4.2: Probability I
Total Duration	9 lessons × 40 minutes = 360 minutes total
Sub-Strand Content	– Experimental probability — conducting repeated trials (coin tosses, dice rolls, picking cards) and computing relative frequencies to estimate probability – Probability space — listing sample spaces using tables, grids, and systematic enumeration for single and combined events – Mutually exclusive events — identifying events that cannot occur simultaneously and applying the relevant probability rules – Independent events — identifying events whose outcomes do not influence each other and applying the relevant probability rules – Addition law of probability — $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, with the special case for mutually exclusive events – Multiplication law of probability — $P(A \cap B) = P(A) \times P(B)$ for independent events, and $P(A \cap B) = P(A) \times P(B\|A)$ for dependent events – Probability tree diagrams — constructing and using tree diagrams to organise outcomes and calculate probabilities of sequential events
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) determine experimental probability by conducting repeated trials and computing relative frequencies in different real-life situations; b) construct and interpret a probability space (sample space) for single and combined events using tables, grids, and systematic listing; c) identify and distinguish between mutually exclusive events and independent events, giving real-life examples from the Kenyan context; d) apply the addition law and multiplication law of probability to solve problems involving combined events; e) construct and use probability tree diagrams to determine probabilities of sequential events and appreciate the role of probability in everyday decision-making.
Core Competencies	– Critical Thinking and Problem Solving: Learners analyse data from experiments, identify patterns in outcomes, and apply probability laws to solve multi-step problems involving real-life scenarios such as weather prediction in Nairobi or crop yield chances in the Rift Valley. – Communication and Collaboration: Learners discuss predictions, share experimental results in groups, and co-construct probability tree diagrams, articulating mathematical reasoning clearly to peers and the teacher. – Creativity and Imagination: Learners design their own probability experiments using locally available materials (coins, bottle tops, seeds) and invent scenarios to model with tree diagrams.

	– Digital Literacy: Learners engage with YouTube demonstrations and Khan Academy simulations to observe probability concepts in action and connect digital representations to pencil-and-paper methods. – Learning to Learn: Learners reflect on how their initial predictions compare with experimental and theoretical results, building metacognitive awareness of uncertainty and mathematical reasoning.
Core Values	– Responsibility: Learners take responsibility for accurate data collection during experiments and honest recording of results, understanding that skewed data leads to misleading probability estimates. – Respect: Learners respect differing predictions and experimental outcomes among peers, recognising that uncertainty is inherent in probability and all contributions to group data are valued. – Integrity: Learners uphold integrity by recording experimental results truthfully, even when outcomes differ from predictions, and by citing sources of data used in probability problems. – Social Justice: Learners explore how probability and risk assessment relate to fair decision-making in society — e.g., equitable resource allocation in Kenyan counties based on likelihood of need. – Unity: Learners collaborate across ability levels in group experiments, appreciating that pooling data from many trials produces more reliable probability estimates — a metaphor for community strength.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners pose investigable questions such as "How likely is it to get two heads when tossing two coins?" and refine questions as they gather experimental evidence. – Planning and Carrying Out Investigations: Learners design and conduct repeated-trial experiments (coin tosses, dice rolls, spinner experiments using locally made spinners) to gather frequency data for estimating experimental probability. – Analysing and Interpreting Data: Learners tabulate results from experiments, compute relative frequencies, compare them to theoretical probabilities, and identify trends as the number of trials increases. – Using Mathematics and Computational Thinking: Learners apply the addition law, multiplication law, and tree diagram methods as formal mathematical tools to compute probabilities precisely. – Constructing Explanations: Learners explain why experimental probability approaches theoretical probability with more trials (Law of Large Numbers) using evidence from their own data. – Obtaining, Evaluating, and Communicating Information: Learners interpret probability presented in real-life Kenyan contexts (weather forecasts, health risk reports, sports statistics) and communicate findings using correct mathematical language and notation.
Pertinent & Contemporary Issues (PCIs)	– Financial Literacy: Learners connect probability to risk assessment in savings, insurance, and investment decisions — for example, estimating the probability of drought in Turkana County affecting crop income and the role of index-based insurance. – Health Education: Learners explore how probability underpins medical testing and disease screening — e.g., the probability that a malaria rapid test gives a correct positive result in a Kenyan clinic. – Environmental Education: Learners apply probability to weather and climate data — e.g., estimating the likelihood of rainfall in Kericho tea farms in any given month based on historical records. – Life Skills: Learners practise decision-making under uncertainty, recognising that understanding probability helps them make informed choices in everyday Kenyan life, from games of chance to career planning. – Citizenship: Learners discuss how governments and county authorities use probability models in public policy — such as estimating the probability of flooding along River Nyando — and the ethical responsibility to act on such data.
Career Connections	– Actuarial Science and Insurance: This sub-strand directly underpins actuarial work, where professionals compute probabilities of risk events (accidents, illness, death) to price insurance products for Kenyan firms such as Jubilee Insurance or AAR. – Medicine and Public Health: Epidemiologists and clinicians at facilities like Kenyatta National Hospital use probability to interpret diagnostic test reliability, model disease spread, and design clinical trials. – Data Science and Statistics: Data analysts at organisations such as the Kenya National Bureau of Statistics (KNBS) use

	probability theory as the foundation for sampling, inference, and predictive modelling. – Meteorology and Environmental Science: Kenya Meteorological Department forecasters apply probability to issue rainfall and drought probability outlooks that guide farmers, relief agencies, and county governments. – Engineering and Quality Control: Engineers at manufacturing plants (e.g., EABL, Bidco) use probability in quality-control sampling to estimate the likelihood of defective products in a production batch. – Finance and Banking: Analysts at Kenyan banks (KCB, Equity Bank) and the Nairobi Securities Exchange use probability models to assess investment risk and price financial derivatives.
Focus for Lessons	This unit introduces Grade 10 learners to the formal study of probability through a carefully sequenced progression — from hands-on experimental probability to the construction of probability spaces, the classification of events, and the application of addition and multiplication laws, culminating in probability tree diagrams. The unit is anchored in an engaging real-life phenomenon drawn from the Kenyan context, ensuring learners see probability not as an abstract mathematical exercise but as a powerful tool for reasoning about uncertainty in everyday life. Throughout the nine lessons, learners move from concrete experimentation and observation toward increasingly abstract and symbolic mathematical reasoning, building both conceptual understanding and procedural fluency.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we use mathematics to measure and predict the likelihood of uncertain everyday events — like Mama Njeri's mandazi sales — so we can make smarter decisions even when we cannot be certain of the outcome? KICD KEY INQUIRY QUESTIONS: 1. How do we determine the probability of an event? 2. How do we use probability to make decisions in real-life situations?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Mandazi Stall Mystery: Why Can't Mama Njeri Predict Her Daily Sales?" Every morning at Gikomba Market in Nairobi, Mama Njeri fries a fresh batch of mandazi to sell. Some days she sells out within two hours and turns away hungry customers; other days she is left with half her batch unsold and loses money. She has kept a notebook for three months recording how many mandazi she sells each day, yet she still cannot perfectly predict tomorrow's sales. When students examine her notebook data, they notice something puzzling: even though the average daily sales are fairly consistent, individual days vary wildly — and the proportion of "good days" (selling at least 80% of her stock) seems to hover around a stable fraction the longer they look at her records. Why does the fraction of good days stabilise over time even though each individual day is unpredictable? How can mathematics help Mama Njeri make a better decision about how many mandazi to fry each morning, even though she can never be certain? This phenomenon surfaces the core tension of the unit: individual outcomes are uncertain, yet patterns of outcomes over many trials are surprisingly predictable — the very foundation of probability.
Storyline Thread	Lesson 1 — INTRODUCING UNCERTAINTY (Experimental Probability I): Students examine Mama Njeri's sales notebook data and make initial predictions about her "good day" fraction. They conduct their own repeated-trial coin and bottle-cap experiments, record results in frequency tables, and compute relative frequencies. They discover that experimental probability stabilises with more trials and update the Driving Question Board (DQB) with their first questions and observations. Lesson 2 — EXPERIMENTAL PROBABILITY II (More Trials, More Patterns): Students pool class data from Lesson 1 experiments to dramatically increase the number of trials, observing how relative frequency converges toward a stable value. They compare their bottle-cap probability (~0.3) with a coin (~0.5), distinguishing equally likely from unequally likely outcomes. They formally define experimental probability as P(E) = (number of favourable outcomes) ÷ (total number of trials) and revisit Mama Njeri's notebook through this lens.

	Lesson 3 — PROBABILITY SPACE (Sample Space & Theoretical Probability): Students move from experimental to theoretical probability by constructing sample spaces for single events (one die, one coin) and combined events (two coins, one coin + one die) using lists, two-way tables, and grids. They define the probability space, the range of probability ($0 \le P(E) \le 1$), and theoretical probability P(E) = (number of favourable outcomes) ÷ (total outcomes in sample space). They connect this to the Kisumu matatu scenario. Lesson 4 — MUTUALLY EXCLUSIVE EVENTS: Students classify pairs of events as mutually exclusive or not using sorting activities and Venn diagrams. They identify that mutually exclusive events cannot occur simultaneously (e.g., a matatu cannot be both red AND blue) and derive $P(A \cup B) = P(A) + P(B)$ for mutually exclusive events. They apply this to solve problems drawn from Kenyan market, sports, and weather contexts. Lesson 5 — INDEPENDENT EVENTS: Students investigate whether the outcome of one event affects another by analysing sequential coin tosses and the Eldoret seed germination scenario. They define independence formally: $P(A \cap B) = P(A) \times P(B)$ and verify this against experimental data. They distinguish independence from mutual exclusivity — a common misconception addressed explicitly. Lesson 6 — ADDITION LAW OF PROBABILITY: Students generalise from mutually exclusive events to the full addition law $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ using Venn diagrams. They apply the law to the Nakuru rainfall puzzle and other combined-event problems. They use the DQB to connect this lesson back to Mama Njeri's problem: on how many days might she experience high sales OR bad weather — two overlapping events? Lesson 7 — MULTIPLICATION LAW OF PROBABILITY: Students derive and apply the multiplication law for independent events $P(A \cap B) = P(A) \times P(B)$ and extend it to dependent events $P(A \cap B) = P(A) \times P(B \mid A)$ using conditional reasoning. They apply the law to multi-step scenarios: the probability that two randomly selected maize seeds (without replacement) both germinate. Lesson 8 — PROBABILITY TREE DIAGRAMS: Students construct tree diagrams to organise sequential outcomes for two- and three-stage experiments, multiplying along branches and adding across branches. They use tree diagrams to revisit the matatu scenario and the seed germination problem, appreciating how tree diagrams make complex combined-event calculations systematic and visual. Lesson 9 — SYNTHESIS & UNIT ASSESSMENT (Helping Mama Njeri): Students apply everything learned to a culminating performance task: using historical sales data, they compute experimental probability, construct the probability space for a simplified model, classify events, apply addition and multiplication laws, and draw a tree diagram to advise Mama Njeri on the optimal number of mandazi to fry. Groups present recommendations with mathematical justification, and the DQB is reviewed to celebrate questions answered and surface remaining curiosities.
Supporting Phenomena	– The Spinning Bottle Cap: Students spin a plastic bottle cap on a desk and record whether it lands "up" or "down" over 50 trials. They find that neither outcome is equally likely — the experimental probability of "up" stabilises around 0.3, not 0.5 — raising the question of how we determine true probability when outcomes are not equally likely. – The Matatu Colour Problem: In Kisumu town, matatus from two competing Saccos (one with red livery, one with blue) arrive at a stage. Students are told that on any given arrival, the chance of a red matatu is 3 in 5. If two matatus arrive in a row, what is the probability that at least one is red? This supports the development of the multiplication law and tree diagrams. – The Rift Valley Rainfall Puzzle: The Kenya Meteorological Department reports that on any day in October in Nakuru, the

	probability of rain is 0.6, and the probability of both rain and strong wind is 0.2. Students ask: what is the probability of rain OR strong wind? This supports the addition law with non-mutually-exclusive events. – The Maize Seed Germination Experiment: A farmer in Eldoret plants 200 maize seeds from two different seed varieties (Hybrid 614 and Hybrid 513) in equal proportions. Records show that Hybrid 614 seeds have a 90% germination rate and Hybrid 513 seeds have a 75% germination rate. If one seed is picked at random, what is the probability it germinates? This supports combined probability and tree diagram reasoning in an agricultural context.

LESSON 1 (80 minutes): The Mandazi Stall Mystery: Introducing Uncertainty and Experimental Probability	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce the concept of probability as a measure of likelihood by anchoring it in the real-world uncertainty of Mama Njeri's mandazi sales at Gikomba Market, and to establish experimental probability through repeated trials.
Knowledge	Students will know that: (1) probability is a measure of the likelihood of an event, expressed as a value between 0 and 1; (2) a random experiment has uncertain outcomes and a defined sample space; (3) experimental probability is calculated as the relative frequency of an event over many trials (number of favourable outcomes ÷ total number of trials); and (4) experimental probability stabilises and approaches a consistent value as the number of trials increases.
Skills	Students will be able to: (1) identify random experiments, sample spaces, and events in everyday Kenyan contexts; (2) record results of repeated experiments in a frequency table; (3) calculate relative frequencies (experimental probabilities) from collected data; and (4) observe and describe the stabilisation trend of experimental probability as trials increase.
Attitudes	Students will appreciate that mathematics is a practical tool for making better decisions under uncertainty, and will show curiosity and open-mindedness when initial predictions are challenged by experimental evidence.
Key Inquiry Question	How do we determine the probability of an event?
Purpose in Storyline	This is the opening lesson of the unit. It launches the anchoring phenomenon — Mama Njeri's unpredictable mandazi sales — and surfaces the central tension of the unit: individual outcomes are uncertain, yet patterns over many trials are surprisingly predictable. Students make their first predictions, gather their first experimental evidence, and open the Driving Question Board (DQB) with their burning questions. Every subsequent lesson will return to and deepen the understanding begun here.
Safety Notes	No significant safety hazards. If bottle caps are used, ensure edges are not sharp. Remind students not to put small objects in their mouths. Coin and bottle-cap experiments should be conducted on flat desk surfaces to prevent objects from falling and causing distraction or injury.

B. LESSON OVERVIEW

In this lesson, students are introduced to the anchoring phenomenon: Mama Njeri, a mandazi seller at Gikomba Market in Nairobi, cannot reliably predict her daily sales despite keeping three months of careful records. Students examine a simplified version of her sales notebook, make initial predictions about the fraction of her 'good days' (days when she sells ≥ 80% of her stock), and then conduct their own repeated-trial experiments using coins and bottle caps to discover how experimental probability is calculated and how it stabilises over many trials. The lesson closes by launching the Driving Question Board (DQB), where students post their first questions and observations about uncertainty and probability. Key vocabulary introduced: random experiment, outcome, sample space, event, frequency, relative frequency, experimental probability.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Students are presented with the	1. Display or distribute the Mama	Think-Pair-Share with a Prediction	Circulate and read student written

📹 **VIDEO:** [Video: Plinko Probability](#) Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	anchoring phenomenon: a short narrative about Mama Njeri's mandazi stall at Gikomba Market, along with a simplified one-week excerpt from her sales notebook (a printed or projected table showing 7 days of data: mandazi fried vs. mandazi sold). Without any instruction, students individually write a prediction in their exercise books: (1) On what fraction of days did Mama Njeri have a 'good day' (selling ≥ 80% of her stock)? (2) If you had to advise Mama Njeri on whether tomorrow will be a good day, what would you say, and why? Students then share predictions with a partner (Think-Pair-Share). The class hears 4–5 predictions; disagreements are noted on the board without resolution — the tension is left open deliberately.	Njeri scenario card and the 7-day notebook excerpt. Read the scenario aloud expressively: 'Every morning Mama Njeri wakes up at 4 a.m., fries her mandazi, and carries them to Gikomba. Some days she sells out. Some days she goes home with half a basket. She has written down every single day for three months — and she still cannot tell you what tomorrow will look like. Why not?' 2. Pause for effect. Say: 'Before I teach you anything, I want to know what YOU think. Look at this one week of her data. In your exercise book, answer these two questions.' [Display the two prediction questions on the board. Allow 4 minutes of silent individual writing.] 3. After 4 minutes, say: 'Turn to your neighbour. You have 2 minutes. Share your fraction estimate and your advice for Mama Njeri.' 4. After pair discussion, cold-call FOUR students (select a mix of confident and quieter voices): 'Akinyi, what fraction did you get and why?' — wait 10 seconds after each student speaks before moving to the next. 5. Record all different estimates on the board in a visible 'Prediction Bank.' Do NOT correct them. Say: 'We have different answers. Some of you said one-third, some said one-half. How will we find out who is closest to right? That is what today's lesson will help us discover.' 6. Explicitly connect to phenomenon: 'This is exactly Mama Njeri's problem. She has guesses — but how do we turn	Bank: Students externalise prior intuitions before instruction, creating cognitive dissonance when predictions differ. Recording all predictions publicly (without judgment) motivates the need for a mathematical method to settle the disagreement — directly motivating the concept of experimental probability.	predictions during the 4-minute silent phase. Observable indicators of readiness: (1) Student can identify at least one 'good day' from the table (sells ≥ 80%); (2) Student attempts a fraction or ratio, even if incorrect. Flag students who write only 'I don't know' for targeted support during the Observe phase. Collect 3–4 exercise books to scan written predictions; return before end of lesson.

		guesses into mathematics?'		
Observe Phase 📹 **VIDEO:** [Video: Plinko Probability] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Part A — Mama Njeri's Full Notebook (10 min): Students receive a printed table summarising 60 days of Mama Njeri's sales data (grouped into sets of 10 days). In pairs, they count the number of 'good days' in each group of 10 and record running totals. They then calculate the running relative frequency (good days so far ÷ total days so far) after every 10 days and plot these values on a pre-drawn number-line graph (x-axis: total days observed; y-axis: relative frequency from 0 to 1). Part B — Coin and Bottle-Cap Experiment (15 min): Each group of 4 students conducts TWO repeated-trial experiments. Experiment 1: Flip a 1-shilling coin 40 times. After every 10 flips, record the cumulative number of heads and compute the running relative frequency of heads. Experiment 2: Toss a metal bottle cap (soda cap) 40 times — note it is NOT a fair coin; it lands 'crown-up' or 'crown-down' with unequal chances. After every 10 tosses, record the cumulative crown-up count and compute relative frequency. Students record all data in a structured frequency table in their exercise books and plot both running relative frequency graphs alongside their Mama Njeri graph.	1. Distribute the 60-day Mama Njeri data table. Say: 'You are now going to be data detectives. Work with your partner. For every 10 days, count the good days and calculate this fraction: good days so far divided by total days so far. Write it as a decimal. Plot it on your graph.' Allow 8 minutes. Circulate — prompt stuck pairs: 'How many good days have you found in the first 10 days? What is that out of 10?' 2. After 8 minutes, pause the class. Cold-call TWO groups to share their relative frequency after day 10 and after day 60. Ask: 'What do you notice about the fraction as the number of days gets bigger?' Wait 12 seconds. Acknowledge: 'Hold that thought — we are about to see if the same thing happens with coins.' 3. Distribute coins and bottle caps to groups. Say: 'You will flip this coin 40 times in your group — one person flips, one records, two check. Switch roles every 10 flips. Ready? Go.' Monitor pacing — groups should complete 40 flips in about 7 minutes. Prompt accurate recording: 'Make sure you are writing the CUMULATIVE total, not just each set of 10.' 4. After coin experiment, direct groups to the bottle-cap experiment. Say: 'Now replace the coin with this bottle cap. It is not a coin — do you think it will behave the same way? We will find out.' Allow 6 minutes. 5. With 2 minutes remaining, say: 'Stop where you are and look at all three graphs in front of you — Mama Njeri's, the coin's, and the	Parallel Data Collection with Running Relative Frequency Graphs: By plotting three simultaneous data sets (Mama Njeri's sales, coin flips, bottle-cap tosses), students directly observe the Law of Large Numbers in action across multiple contexts. The visual convergence of the graphs toward a stable value makes the abstract concept of probability as a limit of relative frequency tangible and cross-contextually grounded.	Check frequency tables as you circulate — look for: (1) Correct cumulative totals (not just per-set counts); (2) Relative frequency expressed as a decimal between 0 and 1; (3) A plotted graph with a visible trend. Common error to watch for: students dividing good days in a set of 10 by 10 (instead of cumulative good days by cumulative total). If observed, pause the group and ask: 'If Mama Njeri had 3 good days in week 1 and 4 good days in week 2, out of all 20 days so far, what is the fraction?' Use this as a teaching moment for the whole class if more than two groups make the same error.

		bottle cap's. What pattern do you see across all three?' Wait 15 seconds of silence. Say: 'Write ONE observation sentence in your book before we discuss.'		
Explain Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students participate in a structured whole-class discussion to formalise the patterns they observed. The teacher introduces key vocabulary (random experiment, outcome, sample space, event, experimental probability) by anchoring each term directly to the Mama Njeri context and the coin/bottle-cap experiments. Students then work in pairs to complete a 'Concept Mapping' task: they are given six terms on sticky notes or strips of paper and must arrange them in a logical map with connecting arrows and phrases. Finally, students individually solve three short application problems: (1) From the 60-day Mama Njeri table, calculate the experimental probability of a good day. (2) A student flipped a coin 50 times and got 23 heads — what is the experimental probability of heads? (3) Kamau rolled a die 30 times in Kisumu and got a '6' exactly 4 times — what is the experimental probability of rolling a 6?	1. Open the discussion: 'Let's name what we saw. Otieno, describe what happened to your relative frequency graph as you did more coin flips.' Cold-call, wait 12 seconds. Affirm the observation, then say: 'Mathematicians call this stabilisation. The value it stabilises toward is what we call the EXPERIMENTAL PROBABILITY.' Write the definition on the board: Experimental Probability = Number of times event occurred ÷ Total number of trials. 2. Anchor each vocabulary term: 'Mama Njeri's morning routine — frying and selling mandazi — is a RANDOM EXPERIMENT because the outcome is uncertain. The set of all possible outcomes — good day or not-good day — is the SAMPLE SPACE. A specific outcome we care about — having a good day — is an EVENT.' Write each term and its Mama Njeri example side by side on the board. 3. Say: 'Now with your partner, take these six terms and build a concept map. Use arrows and write a short phrase on each arrow to show how the terms connect. You have 5 minutes.' [Terms: random experiment, outcome, sample space, event, trial, experimental probability.] Circulate and prompt: 'What comes before an outcome? What is the relationship between event and sample space?' 4. After concept mapping, project	Vocabulary Anchoring + Concept Mapping: Introducing formal vocabulary AFTER students have experienced the concepts through data collection ensures that terms attach to genuine understanding rather than rote memorisation. The concept map externalises relational thinking and reveals misconceptions (e.g., confusing 'outcome' with 'event') before they solidify.	Collect concept maps from three randomly selected pairs. Look for: (1) Correct directional arrows (e.g., 'a trial produces an outcome'); (2) Sample space correctly positioned as the set containing all outcomes; (3) Experimental probability correctly linked to trials and events. For the three application problems, observe student work — target indicator: student correctly places the count of favourable outcomes in the numerator and total trials in the denominator. Students who invert the fraction or use total outcomes (not trials) need re-teaching of the formula before Lesson 2.

		the three application problems. Say: 'Try problem 1 silently for 2 minutes.' Cold-call one student for the answer. Say: 'Wanjiku, walk us through your calculation.' Wait 10 seconds. Then move to problems 2 and 3 with pair discussion. 5. Return to the phenomenon: 'So — using Mama Njeri's 60-day data, we now have a mathematical estimate of her good-day probability. Is this the same as a guarantee? Can she be 100% certain tomorrow will be a good day?' Wait 15 seconds. Accept 2–3 responses. Close with: 'That tension — between what the mathematics predicts and what actually happens — is the mystery we are going to keep investigating.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are introduced to the Driving Question Board — a dedicated space on the classroom wall (or a large sheet of manila paper) that will be maintained throughout the entire unit. Each student writes TWO questions on separate sticky notes (or small pieces of paper): one question about something they do not yet understand about probability, and one question about how probability might help Mama Njeri or someone else they know make a better decision. Students read their questions aloud in a gallery-walk format and physically post them on the DQB, grouping similar questions together. The teacher labels emergent clusters with working category headings. The class identifies ONE 'burning question' they most want to answer — this is starred on the DQB and will be revisited each lesson.	1. Introduce the DQB: 'On this wall, we are going to track our growing understanding across the whole unit. Every question you post here is a good question — there are no wrong questions on this board. By the end of the unit, we should be able to answer most of them.' 2. Distribute sticky notes (or paper strips). Say: 'Write TWO questions. Question one: something about probability you don't understand yet. Question two: a real-life situation — maybe in your home, your market, your school — where you wonder about the chances of something happening. You have 3 minutes. Write in pen so we can all read it.' 3. Play soft background music (optional) while students write. After 3 minutes, say: 'Stand up. Read your questions to the person next to you. Then walk to the	Driving Question Board (DQB) Launch: The DQB is a metacognitive anchoring tool that makes student thinking visible and cumulative. By publicly posting questions, students take ownership of the inquiry, and the teacher gains real-time formative data about misconceptions and curiosity gaps. Revisiting the DQB each lesson creates a visible narrative of growing understanding that motivates continued engagement.	Read all posted questions before the next lesson. Categorise them into: (1) Conceptual questions (what is probability / how is it calculated); (2) Application questions (how does it help decisions); (3) Misconception-revealing questions (e.g., 'if probability is 0.5 does that mean exactly half will happen?'). Use the misconception-revealing questions to plan targeted clarification in Lesson 2. The starred 'burning question' should be referenced at the start of every subsequent lesson.

		board and post your questions. If someone else has a similar question, post yours nearby.' Allow 4 minutes of gallery walk. 4. After posting, scan the board aloud: 'I see a cluster here about how we CALCULATE probability — several of you are asking about that. I see another cluster about whether probability can be WRONG. And I see questions about GAMES and SPORTS — interesting.' Label clusters with sticky-note headers. 5. Ask: 'Which ONE question do you most want this unit to answer? Call it out.' Tally responses by a show of hands and star the winning question. Say: 'We will come back to this board every lesson and update it as we learn more. Today's lesson has already given us some answers — and many more questions.'		
Model Building Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students create their first Probability Model — a hand-drawn annotated diagram in their exercise books titled 'My Model of Mama Njeri's Good-Day Probability.' The model must include: (1) A labelled number line from 0 to 1 showing where they would place the experimental probability of Mama Njeri's good day; (2) Labels for 'impossible' (0), 'certain' (1), and 'equally likely as unlikely' (0.5); (3) An arrow showing where their coin and bottle-cap experimental probabilities fell; (4) A written sentence explaining what the model shows and one thing it does NOT yet tell them. Students share models with their group and agree on one revision or addition before the lesson ends. Students are told	1. Say: 'Scientists and mathematicians build models to represent what they understand — and to show what they still need to figure out. You are going to build your first probability model right now. It does not have to be perfect. It has to be honest about what you know and what you don't.' 2. Display the model requirements on the board. Say: 'You have 6 minutes. Draw and annotate your model individually.' Circulate and prompt: 'Where on this line would impossible sit? Where would certain sit? Where does Mama Njeri's good-day probability go, based on the data?' 3. After 6 minutes, say: 'Show your model to your group. Discuss: what is one thing you could add or	Cumulative Model Building with a Probability Number Line: The number line model is deliberately simple at this stage — it anchors the key idea that probability is a quantity with a range, not just a word ('likely,' 'unlikely'). By requiring students to place their experimental results on the line, the model bridges qualitative language and mathematical representation. The 'what my model does NOT yet tell me' component builds epistemic humility and scientific practice.	Collect 5–6 model pages (photograph or collect exercise books) before students leave. Look for: (1) Number line correctly labelled with 0 and 1 at the extremes; (2) Mama Njeri's good-day probability placed between 0.5 and 1 (based on the 60-day data, which is designed to yield approximately 0.6–0.65); (3) At least one honest 'what I don't know yet' sentence. Students who place the probability outside the 0–1 range, or who write 'nothing' for what they don't know, are flagged for a brief individual check-in at the start of Lesson 2.

	this model will be revisited and improved in every lesson of the unit.	change to make it more accurate?' Allow 3 minutes of group discussion. 4. Ask two volunteers to share their models under the document camera or by holding them up. For each: 'What did your group decide to add or change? Why?' Wait 10 seconds between responses. 5. Close the lesson: 'This model is the beginning of our thinking. Every lesson in this unit, you will come back to this model and update it as you learn more. By Lesson 5, your model should look very different — and much more powerful. Just like Mama Njeri's notebook gets more useful the more data she adds, your model gets more useful the more mathematics you add to it.' 6. Assign exit task: 'Before tomorrow, ask one adult at home — a parent, a shopkeeper, a neighbour — about a decision they make that involves uncertainty. Write down what they said in one sentence. We will use those examples next lesson.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: \n1. PHENOMENON UPTAKE — Did students genuinely engage with Mama Njeri's dilemma, or did it feel like a contrived word problem to them? What evidence (questions asked, energy during discussion, DQB posts) supports your answer? \n2. PREDICTION DIVERSITY — How wide was the spread of student predictions in the Prediction Bank? A wide spread is healthy — it shows genuine prior uncertainty and motivates investigation. A very narrow spread may mean students are guessing socially rather than thinking independently. How could you adjust the predict phase next time? \n3. EXPERIMENTAL DESIGN MANAGEMENT — Did groups complete 40 trials within the time allocated? Were frequency tables recorded accurately? If groups fell behind, consider pre-recording the first 20 coin flips as a worked example and having groups only generate the second 20 independently. \n4. VOCABULARY RETENTION — During the Explain phase, could students use the terms 'sample space,' 'event,' and 'experimental probability' correctly in their own sentences? If not, plan a 5-minute vocabulary warm-up at the start of Lesson 2. \n5. DQB QUALITY — What categories of questions emerged? Were there questions that suggest deep curiosity (e.g., 'why does it stabilise?') or questions that suggest fundamental confusion (e.g., 'what even is probability?')? Use this to sequence the next lesson's emphasis. \n6. MODEL ACCURACY — Were students able to correctly place experimental probabilities on the 0–1 number line? The most common misconception at this stage is placing probability on a scale of 0–100 (percentage thinking) rather than 0–1. Address this explicitly at the start of Lesson 2 if observed.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed three parallel data sets — Mama Njeri's 60-day sales record, coin-flip results, and bottle-cap toss results — and plotted running relative frequencies for all three. They noticed that relative frequencies fluctuate early but appear to stabilise toward a consistent value as the number of trials increases.
What did I learn?	Students learned that probability is a mathematical measure of likelihood ranging from 0 (impossible) to 1 (certain); that a random experiment has an uncertain outcome and a defined sample space; that experimental probability is calculated as the number of times an event occurs divided by the total number of trials; and that experimental probability becomes more reliable (stabilises) as the number of trials increases.
How does this explain the phenomenon?	Students explained the pattern in Mama Njeri's notebook and their own experiments using the concept of experimental probability. They described why individual days are unpredictable yet the fraction of good days stabilises over 60 days — because with more trials, random variation averages out and the relative frequency converges toward the true probability of a good day. This explains why keeping a longer notebook makes Mama Njeri's data more mathematically useful for planning.

LESSON 2 (80 minutes): Experimental Probability II: More Trials, More Patterns	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To deepen students' understanding of experimental probability by pooling large datasets, observing the convergence of relative frequency, and formally applying P(E) = favourable outcomes ÷ total trials to Mama Njeri's mandazi sales data.
Knowledge	Students will know that: (1) experimental probability is defined as P(E) = number of favourable outcomes ÷ total number of trials; (2) relative frequency stabilises (converges) as the number of trials increases; (3) equally likely outcomes (e.g. a fair coin) and unequally likely outcomes (e.g. a bottle-cap) produce different stable probabilities; (4) a larger sample size gives a more reliable estimate of true probability.
Skills	Students will be able to: (1) calculate experimental probability using the formula P(E) = favourable outcomes ÷ total trials; (2) pool class data to increase trial count and compute updated probabilities; (3) construct and interpret a relative-frequency graph showing convergence; (4) distinguish between equally likely and unequally likely outcomes using experimental evidence; (5) apply the formula to Mama Njeri's notebook data to estimate the probability of a 'good day'.
Attitudes	Students will appreciate that: (1) mathematics provides a reliable tool for making decisions under uncertainty; (2) collecting more evidence (more trials) leads to more trustworthy conclusions; (3) patience and systematic record-keeping — like Mama Njeri's notebook — are valuable scientific and mathematical habits.
Key Inquiry Question	How do we determine the probability of an event, and how does increasing the number of trials improve our estimate?
Purpose in Storyline	In Lesson 1, students ran individual experiments with bottle-caps and noticed variability. Lesson 2 pools all class data to reveal that the relative frequency of 'cap lands up' stabilises near ~0.3 over many trials — mirroring exactly why Mama Njeri's fraction of 'good days' stabilises over three months even though individual days are unpredictable. Students leave with the formal definition of experimental probability and a working understanding of convergence.
Safety Notes	No hazardous materials are used. Ensure bottle-caps are clean and free of sharp edges. Students should keep bottle-caps on desk surfaces and not throw them. Coin-flipping should be done seated to avoid disruption.

B. LESSON OVERVIEW

Students begin by re-examining the Lesson 1 bottle-cap data they recorded individually, then systematically pool all class results on a shared tally chart on the board. As the total trial count grows from ~30 (individual) to ~900+ (whole class), students plot a live relative-frequency graph and watch the probability estimate converge toward a stable value. They repeat this comparison with a coin (equally likely outcomes) and contrast the two convergence lines. The teacher formally introduces P(E) = favourable outcomes ÷ total trials, anchoring it to both the bottle-cap experiment and Mama Njeri's three-month notebook. Students calculate the experimental probability of a 'good day' from her records, add a new entry to the Driving Question Board, and update their probability models for the unit phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Students are shown the class's	Display the individual tally sheets	Prediction with Justification — By	Observable indicator: Every

📹 **VIDEO:** [Finding area with fractional sides 2] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	collective Lesson 1 bottle-cap data (all individual tallies listed on the board but not yet summed). They are asked: 'If we combine everyone's results — roughly 900 flips in total — what do you predict the fraction of cap-up results will be? Will it be closer to your own result, or will it settle somewhere different?' Students write a prediction (a fraction or decimal between 0 and 1) in their exercise books along with a one-sentence justification. They also predict whether pooling coin-flip data would give a different converging value than the bottle-cap data, and why.	from Lesson 1 on the board (or projected). Say: 'Before we add anything together, look at these individual results. Some of you got cap-up 8 out of 30 times. Others got 12 out of 30. Take 10 seconds — predict silently: when we pool everything, what single fraction do you think we will land on?' [WAIT TIME: 12 seconds]. Cold-call 4 students: ask each to share their predicted fraction AND their reason. Record the four predictions visibly on the board under the heading 'Our Predictions — We Will Test These.' Then ask: 'Does anyone predict the coin will converge to the same fraction as the bottle-cap? Hands up.' Note the show-of-hands count aloud. Say: 'Keep your predictions in mind — we are about to find out together.'	committing to a written prediction before seeing pooled data, students activate prior knowledge from Lesson 1 and create cognitive readiness for the convergence pattern. Recording predictions publicly on the board creates accountability and a reference point for the 'Explain' phase reveal.	student has a written fraction prediction with a justification sentence in their exercise book before pooling begins. Teacher circulates and checks 6–8 books. Flag students who write 'I don't know' — pair them with a neighbour for a 30-second discussion before cold-calling. Listen for whether students reference their own Lesson 1 data as a basis for their prediction (desired) versus guessing randomly (needs support).
Observe Phase 📹 **VIDEO:** [Finding area with fractional sides 2] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The class builds a shared data table on the board. Each student (or pair) reads out their Lesson 1 bottle-cap tally: total flips and number of cap-up results. A designated class recorder adds each group's data to a running cumulative total. After every five groups contribute, the class pauses, calculates the current cumulative relative frequency (cap-up ÷ total so far), and one student plots this point on a large class relative-frequency graph drawn on the board (x-axis: cumulative trials; y-axis: relative frequency 0–1). Students copy the graph into their books in real time. The process repeats until all groups have contributed. Students then repeat the exercise with coin-flip data from Lesson 1 (if collected) or perform 20 quick coin flips each now, adding coin data to a second	Prepare the board with a large pre-drawn axes grid (x: 0 to 960 trials, y: 0 to 1.0, with a dashed horizontal line at y = 0.5 labelled 'coin benchmark'). Assign one student as 'Data Recorder' and one as 'Graph Plotter.' As groups contribute data say: 'Group 3, read your numbers clearly. Everyone: add these to your running total in your table NOW.' After each plot point, pause and ask the class: 'Is the line settling, or still jumping around a lot?' [WAIT TIME: 10 seconds]. Cold-call 3 different students at three different cumulative totals (~150 trials, ~450 trials, ~900 trials) to describe what they notice about the graph's behaviour. When both lines are complete, say: 'Look at where each line seems to be settling. Bottle-cap — point to it. Coin — point to it. What do you notice	Live Cumulative Graphing — Students construct the convergence graph in real time, data point by data point. This makes the abstract concept of 'relative frequency stabilises over many trials' visually and experientially concrete. The simultaneous comparison of two lines (bottle-cap ~0.3, coin ~0.5) creates a productive contrast that will anchor the distinction between unequally likely and equally likely outcomes in the Explain phase.	Observable indicator: Every student's exercise book contains a completed two-line relative-frequency graph with both axes labelled, both lines plotted, and at least one annotation (e.g. an arrow and the words 'settling here'). Teacher checks 5 books during the graphing activity. Misconception to watch for: students who draw a flat line from the start — they may not understand that early trials are noisy; address by pointing to the jagged left portion of the graph on the board and asking 'Was the line flat here at the beginning?'

	line on the same graph. Students observe and annotate both convergence lines.	about the difference?' [WAIT TIME: 12 seconds]. Cold-call 3 students. Ensure the bottle-cap line visibly settles near 0.3 and the coin line near 0.5.		
Explain Phase 📹 **VIDEO:** [Finding area with fractional sides 2] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher formally introduces the definition and formula for experimental probability: P(E) = number of favourable outcomes ÷ total number of trials. Students apply the formula to three worked examples: (1) using the pooled class bottle-cap data; (2) using the pooled coin data; (3) using Mama Njeri's notebook — she recorded 90 days of data and had 27 'good days' (selling ≥ 80% of stock). Students calculate P(good day) = 27 ÷ 90 = 0.3, then discuss what this means for her decision-making. The teacher introduces the vocabulary: equally likely outcomes (coin), unequally likely outcomes (bottle-cap), and explains that experimental probability is an estimate that improves with more trials. Students write a formal definition and the three worked examples in their notes.	Write on the board in large print: 'P(E) = number of favourable outcomes ÷ total number of trials.' Say: 'This is the formal definition of experimental probability. Say it with me.' [Class reads aloud together once.] 'Now let us use it.' Work through Example 1 (bottle-cap) with full class participation: 'Our total trials were [class total]. Cap-up count was [class total cap-up]. Someone calculate P(cap up) for me — you have 20 seconds, write it in your book.' [WAIT TIME: 20 seconds]. Cold-call 2 students to give the answer and one to explain the division. Repeat for Example 2 (coin). Then say: 'Now — Mama Njeri. She has 90 days of data. 27 were good days. What is P(good day)?' [WAIT TIME: 15 seconds]. Cold-call 3 students. Then ask: 'What does this 0.3 tell Mama Njeri about tomorrow?' [WAIT TIME: 12 seconds]. Cold-call 4 students. Push deeper: 'Is 0.3 a guarantee? Or something else?' Introduce the terms 'equally likely' (coin, P = 0.5) and 'unequally likely' (bottle-cap/mandazi sales, P ≠ 0.5) explicitly, writing both terms on the board with examples.	Worked Example Progression with Phenomenon Anchoring — Moving from a physical object (bottle-cap) to a fair coin to Mama Njeri's real data keeps the abstraction grounded. The three-example structure builds procedural fluency (applying the formula) while the Mama Njeri example reconnects the formula directly to the anchoring phenomenon, ensuring students see the formula as a decision-making tool, not just a calculation.	Observable indicator: Students can independently calculate P(good day) = 27 ÷ 90 = 0.3 and state in one sentence what it means for Mama Njeri's decision. Exit check — after the Mama Njeri calculation, teacher says: 'Write in your own words: what does P = 0.3 mean for Mama Njeri? You have 90 seconds.' Collect 6 random books and read responses. Red flag: students who write 'she will have a good day 0.3 times' without connecting it to 30% of days over many days — address with a whole-class clarification before moving on.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Finding area with fractional sides 2 Source: Khan	Students return to the class Driving Question Board. They review the two KICD Key Inquiry Questions and discuss in pairs for 2 minutes: 'What new evidence from today can we add to help answer each question?' Each pair writes one claim on a sticky note	Point to the DQB and say: 'In Lesson 1, we put questions on this board. Today we have evidence to start answering them. Work with your partner — 2 minutes — write one claim that our data today supports.' [WAIT TIME: 2 full minutes of pair work]. Circulate	Claim–Evidence Linking on the DQB — Requiring students to write in the format 'We now know… because our data showed…' forces explicit connection between empirical evidence (the pooled graph, Mama Njeri's 90-day data) and	Observable indicator: Each pair produces a sticky note with a claim that references specific data (a number, a graph feature, or Mama Njeri's scenario) — not a vague statement like 'probability is useful.' Teacher reads all sticky notes after class and identifies any

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(or index card) in the format: 'We now know that… because our data showed…' and adds it to the DQB under the relevant question. Students also update their personal 'Probability Scale' strip (created in Lesson 1) by marking where P(good day for Mama Njeri) = 0.3 falls, and annotating: 'experimental probability from 90 days of data.' The class briefly shares 3–4 sticky notes aloud, and the teacher highlights any notes that show the connection between more trials → more reliable probability estimate.	and prompt pairs who are stuck: 'What did the graph show us when we added more and more trials?' After 2 minutes, cold-call 4 pairs to read their sticky notes aloud. As each note is placed on the board, ask the class: 'Do you agree this answers part of our driving question? Thumbs up / thumbs sideways / thumbs down.' Ensure at least one note explicitly references Mama Njeri and one references the convergence behaviour. Say: 'Notice — we are not finished answering the driving question. What do we still not know?' [WAIT TIME: 10 seconds]. Cold-call 2 students to name a remaining gap — these become 'wonder questions' added to the board in a different colour.	conceptual claims (probability converges, P(E) formula). The public display creates a cumulative, visible record of the class's growing understanding across all 9 lessons.	that show lingering confusion (e.g. conflating experimental probability with theoretical probability) to address at the start of Lesson 3.
Model Building Phase 📹 **VIDEO:** Finding area with fractional sides 2 Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their probability model from Lesson 1 (a diagram or written explanation of why Mama Njeri cannot predict her daily sales). They now revise and extend it in light of today's learning. The model must now include: (1) the P(E) formula labelled; (2) an arrow or annotation showing 'more trials → more stable estimate'; (3) Mama Njeri's P(good day) = 0.3 marked on a 0–1 probability scale within the model; (4) a distinction between 'individual day is unpredictable' and 'pattern over many days is predictable.' Students annotate in a different colour so revision is visible. In the final 3 minutes, two volunteers share their updated models with the class and explain one change they made and why.	Say: 'Take out the model you drew in Lesson 1. You have 8 minutes to revise it using what you learned today. Your model must show the formula, the convergence idea, and Mama Njeri's 0.3. Use a different colour pen for today's additions so we can see your thinking grow.' [WAIT TIME: 8 minutes of individual model work — circulate continuously]. Prompt students who have not yet included convergence: 'Your graph today showed the line was jumpy at first, then settled — can your model show that?' After 8 minutes, cold-call 2 volunteers to display and explain their models. Ask the class after each: 'What did this model show well? What could make it even clearer?' [WAIT TIME: 10 seconds each time]. Close by saying: 'In Lesson 3, we will ask: what if we never ran the experiment — can mathematics predict probability without any	Annotated Model Revision — Using a different colour for Lesson 2 additions makes students' conceptual growth physically visible, to themselves and to the teacher. The requirement to include both the formula and the convergence idea forces integration of procedural knowledge (P(E) formula) with conceptual knowledge (why more trials matter), which is essential for transferring understanding to new contexts like Mama Njeri's decision.	Observable indicator: Updated model contains all four required elements (formula, convergence annotation, P = 0.3 on scale, individual vs. pattern distinction) in a visibly different colour from Lesson 1 work. Teacher collects 8 models at the end of class for detailed review. Success threshold: at least 6 of 8 collected models include the convergence annotation and Mama Njeri's 0.3. Models missing these elements flag students who need a brief targeted intervention at the start of Lesson 3.

		trials at all? Keep that question in mind.'		

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Did the pooled data graph clearly show convergence — was the visual convincing enough that students could see the line 'settling'? If the class had fewer than 15 groups, the convergence may have been weak; consider supplementing with a pre-prepared dataset of 500 simulated bottle-cap flips next time. (2) Were students able to correctly apply P(E) = favourable ÷ total to Mama Njeri's 90-day data independently, or did most need the worked example on the board? If fewer than 70% solved it independently, introduce a structured calculation scaffold (a table template) in Lesson 3. (3) Did the distinction between equally likely (coin) and unequally likely (bottle-cap) land clearly? Check the DQB sticky notes — if none mention this distinction, explicitly revisit it as a warm-up in Lesson 3. (4) Were the 8 collected models sufficient quality to reveal misconceptions before Lesson 3? Identify any students who conflated experimental probability with certainty (e.g. 'Mama Njeri will have 27 good days next month') — these students need a targeted question at the start of Lesson 3: 'Does P = 0.3 mean exactly 27 out of the next 90 days will be good?' (5) Was 80 minutes sufficient? The live graphing activity in Phase 2 is the most time-consuming — if it overran, consider pre-plotting the first 5 data points before class to save 5–8 minutes.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students pooled all Lesson 1 bottle-cap flip data (approx. 900 total trials) onto a shared class tally chart and constructed a live cumulative relative-frequency graph on the board, plotting how the fraction of 'cap-up' results changed as more trials were added. They also plotted coin-flip data on the same graph for comparison, producing two visible convergence lines: bottle-cap settling near 0.3 and coin settling near 0.5.
What did I learn?	Students formally learned that experimental probability is defined as P(E) = number of favourable outcomes ÷ total number of trials. They learned that relative frequency converges toward a stable value as the number of trials increases (the Law of Large Numbers at an intuitive level), that equally likely outcomes (fair coin, P = 0.5) differ from unequally likely outcomes (bottle-cap, P ≈ 0.3), and that larger samples produce more reliable probability estimates.
How does this explain the phenomenon?	Students applied the P(E) formula to Mama Njeri's three-month notebook, calculating P(good day) = 27 ÷ 90 = 0.3. They explained that this means approximately 30% of days are 'good days' over many days of trading — not that any specific tomorrow is guaranteed. They connected the convergence pattern in their bottle-cap graph to why Mama Njeri's fraction of good days stabilised over three months even though individual days remained unpredictable, advancing the class's answer to the driving question.

LESSON 3 (40 minutes): Probability Space: Sample Spaces and Theoretical Probability	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of this lesson, students can construct sample spaces for single and combined events and use them to calculate theoretical probability, connecting mathematical structure to real decisions under uncertainty.
Knowledge	Students will know: (1) the definition of a probability space and sample space; (2) that theoretical probability is P(E) = number of favourable outcomes divided by total outcomes in the sample space; (3) that the range of any probability is 0 less than or equal to P(E) less than or equal to 1; (4) how to represent sample spaces using lists, two-way tables, and grids for single events and combined events.
Skills	Students will be able to: (1) list all elements of a sample space systematically; (2) construct two-way tables and grids for combined events such as two coins or one coin and one die; (3) calculate theoretical probability for specified events; (4) compare theoretical probability values to the experimental probability estimates obtained in Lessons 1 and 2.
Attitudes	Students will appreciate that mathematics can impose orderly structure on apparently chaotic situations, building confidence that uncertainty can be reasoned about rather than simply feared.
Key Inquiry Question	How do we determine the probability of an event? This lesson answers this by moving beyond counting trials to reasoning systematically about all possible outcomes before any experiment is run.
Purpose in Storyline	Lessons 1 and 2 showed students that experimental probability stabilises with more trials. Lesson 3 reveals why: there is an underlying theoretical structure — the sample space — that the relative frequency is converging toward. This bridges experimental observation to mathematical theory and equips students with the tool needed for Lessons 4 through 8 where they manipulate probabilities algebraically.
Safety Notes	No safety concerns. Physical materials are limited to coins and dice. Remind students not to throw dice across the room; dice should be rolled gently on desks.

B. LESSON OVERVIEW

Students begin by predicting the probability that a matatu in Kisumu is assigned a route number that is even, using only reasoning rather than experiment. They then observe how a systematically listed sample space confirms or refutes their prediction. Through guided construction of sample spaces (list for one die, two-way table for two coins, grid for one coin plus one die), students build the concept of a probability space and derive theoretical probability. The lesson closes by connecting the P = 0.5 theoretical value for a fair coin to the convergence seen in Lesson 2, and students update the Driving Question Board with a key new insight: Mama Njeri can reason about probability even before she opens her stall.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Finding area with fractional sides 2]	Students are shown a short scenario on the board: Kisumu Matatu Authority assigns each matatu a route number from 1 to	Teacher displays the scenario and says: 'Before we do any experiment at all, I want you to predict — using only your thinking	Anchoring prediction to a concrete Kenyan context (Kisumu matatu route numbers) activates prior numerical reasoning while creating	Teacher scans half-sheets for quality of justification. Look for: students who simply guess versus students who begin counting even

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	12. A matatu is selected at random. Students individually write on a half-sheet: (a) their prediction of the probability that the selected matatu has an even route number, and (b) a one-sentence justification. They then share predictions with a partner before the teacher takes a whole-class show of hands for predictions above 0.5, equal to 0.5, and below 0.5.	— what is the chance the matatu gets an even number. Write your prediction and tell me why you think so.' WAIT TIME: 90 seconds of silent individual writing before pair discussion. After pair talk, teacher asks three volunteers to share reasoning, then records predictions as a tally on the board under three columns: ABOVE half, EXACTLY half, BELOW half. Teacher does NOT confirm the correct answer yet, saying: 'Hold those predictions. We are going to build something that will let us check them precisely.'	productive cognitive dissonance. Recording class predictions publicly creates accountability and motivation to resolve the question.	numbers. If fewer than half the class mentions counting, teacher prompts: 'How many even numbers are between 1 and 12? How many numbers are there in total?' This signals whether students are ready to think about ratios of outcomes.
Observe Phase 📹 **VIDEO:** Finding area with fractional sides 2 Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students work in groups of three. Each group receives a task card with three sub-tasks. Task 1: List all outcomes when one fair die is rolled. Task 2: Complete a partially filled two-way table showing all outcomes when two coins are tossed simultaneously (heads/tails for Coin A across the top, heads/tails for Coin B down the side). Task 3: Complete a grid showing all outcomes when one coin is tossed and one die is rolled (coin outcomes on one axis, die outcomes 1 to 6 on the other). Groups record their completed tables and count total outcomes and favourable outcomes for an assigned event printed on their card (e.g., Group A: both coins show heads; Group B: die shows a multiple of 3; Group C: coin shows tails and die shows an odd number). After 7 minutes, groups share their total outcome counts with the class by calling out numbers while teacher records on the board.	Teacher circulates actively during the 7 minutes. At each group, teacher asks: 'How are you making sure you have not missed any outcome?' and 'How do you know you have not repeated any outcome?' WAIT TIME: After a group gives an answer, teacher waits 5 seconds before responding to allow self-correction. When recording group totals on the board, teacher asks: 'Does every group agree on the total number of outcomes for two coins? Hold up one finger if yes, two fingers if no.' If disagreement exists, teacher invites the differing group to show their table on the board. Teacher then returns to the matatu prediction: 'Now look at the 1-to-12 list from our prediction. How many outcomes are there in total? How many are even?'	Parallel construction of three representations (list, two-way table, grid) allows students to see that the same mathematical object — a sample space — can be displayed in multiple equivalent ways. The varied group tasks ensure every student encounters at least one combined-event structure. Returning to the matatu context ties the new tool back to the opening prediction.	Teacher checks two-way tables for completeness (should have exactly 4 cells for two coins: HH, HT, TH, TT) and grids for combined events (should have exactly 12 cells for one coin and one die). Common error: students list HT and TH as the same outcome. Teacher addresses this by naming the coins: 'Call this one the Nairobi coin and this one the Mombasa coin. Now are HT and TH the same?' Listen for the moment students articulate that order of the coins matters because the coins are distinguishable.
Explain Phase 📹 **VIDEO:**	Teacher leads a whole-class discussion that formalises the	Teacher writes the formula slowly while narrating: 'The top of this	The explicit comparison between the Lesson 2 experimental coin	Exit check: teacher asks three targeted questions in sequence

[Finding area with fractional sides 2](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	vocabulary and the formula. Students copy a structured summary into their notebooks with three labelled sections. Section 1 — Probability Space: the set of all possible outcomes of an experiment; students write the sample space for rolling one die as S = {1, 2, 3, 4, 5, 6} and note that S has 6 elements. Section 2 — Range of probability: teacher writes 0 less than or equal to P(E) less than or equal to 1 on the board and students generate examples of impossible events (P = 0) and certain events (P = 1) from the die and coin contexts. Section 3 — Theoretical probability formula: P(E) = number of favourable outcomes divided by total number of outcomes in the sample space. Students then use this formula to calculate three probabilities from their group tasks, confirm the matatu even-number probability (6 out of 12 = 0.5), and compare P(theoretical coin heads) = 0.5 with the experimental values from Lessons 1 and 2 recorded in their books. Teacher asks: 'In Lesson 2 we saw the relative frequency of heads getting closer and closer to what value as we added more trials? And what does our theoretical calculation give us for a fair coin?'	fraction counts only the outcomes we want — the favourable ones. The bottom counts every outcome that could possibly happen — nothing left out, nothing repeated. This fraction is what we call the theoretical probability.' Teacher then asks: 'Give me an example of an event with probability zero using our die.' WAIT TIME: 10 seconds. 'Give me an example of an event with probability one.' WAIT TIME: 10 seconds. Teacher explicitly connects to the phenomenon: 'Think about Mama Njeri. If her sales could only ever be in the set of whole numbers from 0 to 200, that set of all possible daily sales figures is her probability space. We cannot yet calculate her theoretical probability easily — her situation is more complex — but we now have the concept.' Teacher writes on the DQB anchor strip: 'Theoretical probability tells us what to expect if all outcomes are equally likely.'	result and the Lesson 3 theoretical coin value creates a conceptual bridge that fulfils the storyline promise: experimental probability converges to theoretical probability. Naming the formula components aloud while writing prevents students from memorising symbols without understanding referents.	with hands-up responses. Q1: 'What is the total number of outcomes in the sample space for two coins?' (Expected: 4). Q2: 'What is the probability of getting exactly one head when two coins are tossed?' (Expected: 2 out of 4 = 0.5). Q3: 'Can a probability ever equal 1.3? Explain.' (Expected: No — probability cannot exceed 1.) Teacher notes any student who gives 0.5 for Q1 — a persistent confusion between probability and count of outcomes.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Finding area with fractional sides 2](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#)	Each student receives one sticky note. They respond to the prompt written on the board: 'What new question does today's lesson raise for you about Mama Njeri or about probability?' Students write individually for 2 minutes, then the class moves to the DQB. Teacher reads out five or six notes and clusters them under two emerging headers: QUESTIONS ABOUT	Teacher facilitates the sticky note placement and reads selected notes aloud, asking the class: 'Does anyone else have a question similar to this one? Raise your hand.' Teacher draws a star next to any note that anticipates Lessons 4 or 5 content (e.g., questions about two events happening together, or events that cannot both happen). Teacher	The DQB ritual consolidates metacognitive awareness: students see their own intellectual journey from Lesson 1 through Lesson 3. Clustering questions by theme previews the upcoming lesson arc without revealing content, maintaining curiosity.	Teacher reads DQB notes between lessons to identify misconceptions to address in Lesson 4. Red-flag notes include: 'Probability only works if the outcomes are equally likely' (needs refinement), 'Theoretical probability is always more accurate than experimental' (oversimplification), and 'Sample space is just a list of numbers' (too

📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	MAMA NJERI'S PROBABILITY SPACE and QUESTIONS ABOUT WHAT HAPPENS WHEN EVENTS COMBINE. Students observe that the DQB is becoming richer — some Lesson 1 questions are now answered (why does the fraction stabilise?) while new questions have appeared (what if two things happen at once?).	says: 'Keep those starred questions in mind — we will be answering them very soon.' Teacher updates the class anchor chart with the sentence: 'The sample space gives us all possible outcomes; theoretical probability measures the fraction of those outcomes that are favourable.'		narrow). These inform the opening of Lesson 4.
Model Building Phase 📹 **VIDEO:** [Finding area with fractional sides 2] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students individually complete a model-building task in their notebooks. The task is titled: 'Simplified Matatu Model.' A matatu company in Kisumu operates 3 routes: Route Dunga, Route Kondele, and Route Mamboleo. Each morning, a driver is randomly assigned to one of these routes and simultaneously assigned a morning shift start time chosen at random from two options: 6 a.m. or 8 a.m. Students must: (1) list the complete sample space as ordered pairs (route, time); (2) draw the two-way table; (3) calculate P(driver gets Route Kondele); (4) calculate P(driver starts at 6 a.m.); (5) calculate P(driver gets Route Kondele AND starts at 6 a.m.); (6) write one sentence connecting their answer to how a matatu company manager might use this information to plan staffing. Students who complete early are challenged to redesign the scenario with 4 routes and 3 shift times and recalculate.	Teacher announces: 'You have 6 minutes. Work silently and independently — this is your chance to show yourself what you understand.' Teacher circulates and targets students who struggle to form ordered pairs, whispering: 'Write the route first, then the time. Now do that for every combination.' For students who complete tasks 1 through 5 quickly, teacher poses the extension verbally rather than writing it on the board to avoid distracting others. In the final 2 minutes, teacher asks one volunteer to share their sample space by reading pairs aloud while classmates check their own lists. Teacher corrects silently if a pair is missing by saying: 'We have 6 elements in this sample space — does everyone see 6 pairs?' Teacher closes by returning to Mama Njeri: 'Mama Njeri has more complex outcomes than our matatu model — but the principle is identical. Every possible day of sales belongs to her probability space, and we can begin to assign probabilities to events within it.'	The model-building task is deliberately simpler than the phenomenon (3 routes, 2 times) so students experience success with the new framework. The final sentence connecting to staffing decisions ensures students see theoretical probability as a practical management tool, not an abstract exercise. Ending with a return to Mama Njeri closes the lesson loop and maintains narrative coherence across the unit.	Teacher collects notebooks at the end of class or photographs notebook pages with a phone. Check: (1) sample space has exactly 6 ordered pairs; (2) two-way table has 6 cells with no repeats; (3) P(Kondele) = 1/3; (4) P(6 a.m.) = 1/2; (5) P(Kondele AND 6 a.m.) = 1/6; (6) sentence shows understanding that probability informs decisions. Students scoring below 3 out of 5 on items 1 through 5 receive a targeted prompt card at the start of Lesson 4 reminding them of the sample space listing strategy.

D. TEACHER REFLECTION

After the lesson, consider: (1) Could students distinguish a sample space (all outcomes) from an event (a subset of outcomes) — did they confuse the two when reading

tasks? (2) Did the shift from experimental to theoretical probability feel abrupt for some students — would a side-by-side comparison table of Lesson 2 data versus Lesson 3 calculations help? (3) Were the three representations (list, two-way table, grid) genuinely redundant for all students, or did some students only understand one of them — if so, which representation needs more time in future iterations? (4) Did the Kisumu matatu context feel authentic and engaging, or did students treat it as generic? Consider inviting a student who uses matatus daily to describe how routes are assigned. (5) Were DQB notes rich enough to confirm that students are genuinely curious about combined events, which is the pivot point for Lessons 4 through 8?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that when we list all outcomes of an experiment systematically — using a list, a two-way table, or a grid — we always get the same total count of outcomes regardless of representation. We also observed that the theoretical probability of getting heads on a fair coin (0.5) matched the value our experimental relative frequency was converging toward in Lesson 2.
What did I learn?	We learned that the sample space is the set of all possible outcomes of an experiment, that theoretical probability P(E) equals the number of favourable outcomes divided by the total number of outcomes in the sample space, and that probability is always between 0 and 1 inclusive. We learned that combined events such as tossing two coins require organised tools like two-way tables to avoid missing or repeating outcomes.
How does this explain the phenomenon?	The stabilisation of relative frequency in Lessons 1 and 2 is explained by theoretical probability: as the number of trials grows, the experimental proportion gets closer to the true fraction of favourable outcomes within the sample space. Mama Njeri experiences predictable long-run patterns precisely because there is an underlying probability space governing her sales outcomes, even though individual days remain uncertain.

LESSON 4 (80 minutes): Mutually Exclusive Events: When Two Things Cannot Happen at Once	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to identify mutually exclusive events, classify pairs of events using Venn diagrams, and apply the addition rule $P(A \cup B) = P(A) + P(B)$ to solve real-life probability problems set in Kenyan contexts.
Knowledge	Learners will know that mutually exclusive events cannot occur at the same time, that their intersection is the empty set ($P(A \cap B) = 0$), and that the probability of either mutually exclusive event occurring is found by adding their individual probabilities: $P(A \cup B) = P(A) + P(B)$.
Skills	Learners will be able to: (1) classify pairs of events as mutually exclusive or not mutually exclusive using sorting activities and Venn diagrams; (2) calculate $P(A \cup B)$ for mutually exclusive events; (3) communicate reasoning about event classification using precise mathematical language.
Attitudes	Learners will appreciate that careful mathematical classification of outcomes helps reduce guesswork in everyday decisions — just as Mama Njeri can use probability to make smarter stocking decisions at her mandazi stall.
Key Inquiry Question	How do we determine the probability of an event? How do we use probability to make decisions in real-life situations?
Purpose in Storyline	In previous lessons students discovered that probability measures likelihood (Lesson 1), explored sample spaces and events (Lesson 2), and calculated basic probabilities from Mama Njeri's sales data (Lesson 3). Now in Lesson 4 they investigate a new layer: some pairs of events at the market are mutually exclusive (a mandazi batch cannot be both 'sold out' AND 'half unsold' on the same day), while others are not. Mastering this distinction unlocks the addition rule, moving students closer to building a complete probability model that can advise Mama Njeri on her daily stocking decision.
Safety Notes	No physical hazards. Ensure all printed sorting cards are collected and recycled after the lesson. If using scissors to cut Venn diagram templates, remind learners to handle them carefully and pass scissors handle-first.

B. LESSON OVERVIEW

Learners begin by predicting which pairs of events from Mama Njeri's market can happen on the same day and which cannot, surfacing prior intuitions about mutual exclusivity. They then gather evidence through a hands-on card-sorting activity and Venn diagram construction that makes the structure of mutually exclusive events visible. A guided class discussion formalises the definition and derives the addition rule $P(A \cup B) = P(A) + P(B)$. Learners apply the rule to solve problems drawn from Kenyan market, matatu, sports, and weather contexts. The lesson closes with learners updating the Driving Question Board and revising their cumulative probability model for Mama Njeri's stall.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Mutually Exclusive Events]	Learners receive a set of six event-pair cards drawn from Mama Njeri's world and the wider Kenyan context (e.g., 'Mama Njeri sells out	Distribute the pre-printed event-pair cards (or read them aloud and ask learners to copy into their books). Say: 'Look at each pair of	Predict-then-Discuss (Elicit Prior Knowledge): By sorting before any instruction, learners externalise their intuitive understanding of	Observable indicator: Each learner's written sort (2 columns + justifications) is collected at the end of Phase 1. Teacher scans

Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Probability of Independent Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	her mandazi' AND 'Mama Njeri has mandazi left over at closing time'; 'It rains in Nairobi this afternoon' AND 'It is sunny all afternoon in Nairobi'; 'A matatu is painted red' AND 'The same matatu is painted blue'; 'A customer buys a beef mandazi' AND 'A customer buys a coconut mandazi'). Working individually in silence (3 minutes), learners sort the cards into two columns in their exercise books: CAN happen at the same time vs. CANNOT happen at the same time. They write a one-sentence justification for each decision. Pairs then compare their sorting and note any disagreements (2 minutes).	events. Think about Mama Njeri's stall, matatus on Ngong Road, the sky over Kisumu — can BOTH events in the pair happen at exactly the same time, or does one prevent the other? Sort them silently first.' Allow 3 minutes of individual think time — do NOT accept hands or answers during this window (enforce WAIT TIME). After pair-share, cold-call 3 learners: 'Achieng, which pair did you and your partner disagree on, and what was each of your reasons?' Do not confirm or correct yet — record disagreements on the board as 'Tensions to Resolve.'	events that 'block' each other. Recording disagreements as 'Tensions to Resolve' creates cognitive dissonance that motivates the evidence-gathering phase and keeps the phenomenon central — Mama Njeri's stall provides the anchor context for every card.	for: (1) correct identification of sold-out vs. left-over as mutually exclusive; (2) misconceptions such as treating 'rain' and 'sunshine' as always mutually exclusive in all weather contexts. Flag 2–3 misconception types to address explicitly in Phase 3.
Observe Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Probability of Independent Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of four. Each group receives: (a) a large A3 Venn diagram template (two overlapping circles labelled Event A and Event B, inside a rectangle labelled Sample Space S); (b) a set of 20 small outcome cards representing the sample space of a combined scenario — 'Mama Njeri's morning: type of mandazi batch outcome AND weather in Gikomba Market.' Outcomes include combinations such as (Sold out, Rainy), (Sold out, Sunny), (Half unsold, Rainy), (Half unsold, Sunny), (Sold out, Cloudy), etc. Groups physically place each outcome card inside the correct region of the Venn diagram for two assigned event pairs — one mutually exclusive pair and one non-mutually exclusive pair. They record: (1) What is in the intersection? (2) What does the overlap mean? (3) Can both events occur for the same outcome card? Groups rotate their Venn diagrams to an adjacent	Circulate continuously. At each group, ask probing questions: 'Show me one outcome card that belongs in BOTH circles — does such a card exist for this pair?' and 'What does an EMPTY intersection tell us?' (Allow 10-second WAIT TIME before prompting further.) After gallery rotation, bring the class together. Display a large class Venn diagram. Cold-call 2 group reporters: 'Kamau, your group had the sold-out vs. half-unsold pair — what did your intersection look like, and what did that tell you?' Then cold-call a second group: 'Fatuma, your group had rain vs. sunshine — same question.' Record key observations verbatim on the board beneath the 'Tensions to Resolve' list from Phase 1.	Concrete Manipulative Venn Diagrams (NGSS Practice 2 — Developing and Using Models): Physically placing outcome cards into diagram regions makes the abstract concept of intersection tangible. An empty intersection is visible and memorable — no card can sit in two places at once if the events are mutually exclusive. The gallery-check rotation embeds peer critique (NGSS Practice 7 — Engaging in Argument from Evidence) and surfaces errors before formalisation.	Observable indicator: Teacher checks each group's Venn diagram for correct placement of outcome cards — specifically, whether the intersection region is empty for the mutually exclusive pair and correctly populated for the non-mutually exclusive pair. Misconception to watch: learners who place 'Sold out' AND 'Half unsold' in the intersection because 'some days she almost sells out' — redirect by asking 'Can one day be BOTH fully sold out AND half unsold at closing time?'

	group for a 'gallery check' (2 minutes) and sticky-note any corrections.			
Explain Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Probability of Independent Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Whole-class formalisation. Teacher leads a structured discussion to derive the definition and addition rule. Learners copy the formal definition into their notebooks: 'Two events A and B are mutually exclusive if they cannot both occur in the same trial — their intersection is empty: $A \cap B = \varnothing$, so $P(A \cap B) = 0$.' Teacher then poses: 'If Mama Njeri's batch is either sold out (probability 0.55 based on her 3-month notebook) or half-unsold (probability 0.30), what is the probability that she experiences one OR the other outcome on a given day?' Learners attempt the calculation individually (2 minutes), then share with a partner. Teacher formalises: $P(A \cup B) = P(A) + P(B) = 0.55 + 0.30 = 0.85$ — only valid because the events are mutually exclusive. Learners then practise three tiered problems: (1) Easy — A bag of maize at Wakulima Market is chosen at random; P(grade A) = 0.4, P(grade B) = 0.35 — find P(grade A or grade B). (2) Medium — In a school sports day in Eldoret, P(class wins relay) = 0.25, P(class wins javelin) = 0.40, P(class wins both) = 0 — find P(class wins at least one event). (3) Challenge — A weather forecast for Mombasa Coast gives P(heavy rain) = 0.3, P(light drizzle) = 0.25, P(sunny) = 0.45; are all three pairs mutually exclusive? Verify that probabilities sum correctly and explain why. Learners self-mark using an answer strip provided after all three problems are attempted.	Open the formalisation with: 'Let's resolve the tensions on our board using what we saw in the Venn diagrams.' Point to the empty intersection from Phase 2. Say: 'Mathematicians gave this a name — mutually exclusive events.' Write the formal definition slowly, asking learners to read it back chorally. For the Mama Njeri calculation, say: 'I want EVERYONE to try this alone first — no talking for 2 minutes.' (Enforce silence and WAIT TIME.) After partner share, cold-call 3 learners at random: 'Otieno, what answer did you get and HOW did you get it?' After the three-problem practice, circulate and look for learners adding probabilities for non-mutually exclusive events on problem 3 — pull one such paper (anonymously) to display and ask: 'What question should we ask BEFORE we add two probabilities together?' Elicit: 'Are the events mutually exclusive?'	Gradual Formalisation with Tiered Practice (NGSS Practice 6 — Constructing Explanations): Moving from the concrete Venn diagram evidence to symbolic notation ($P(A \cap B) = 0 \rightarrow P(A \cup B) = P(A) + P(B)$) respects the learning progression. Tiered problems ensure all learners access the rule while challenge problems push higher-order thinking. Connecting every problem back to a Kenyan context (mandazi, maize grades, Eldoret sports day, Mombasa weather) maintains coherence with the anchoring phenomenon.	Observable indicator: Learner responses to the three practice problems — teacher checks that (1) all learners correctly add for problems 1 and 2, (2) learners on problem 3 identify that heavy rain and light drizzle may not be mutually exclusive in all definitions and that probabilities summing to 1.0 is consistent with the sample space. Exit checkpoint: ask learners to hold up one finger if they got all three correct, two fingers if they got two, etc. — use this to group learners for the DQB phase.

Driving Question Board (DQB) Creation 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Probability of Independent Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board displayed at the front. The DQB carries the anchoring question: 'How can mathematics help Mama Njeri make smarter decisions about her mandazi stall?' Each learner receives two sticky notes. On the GREEN note they write one thing today's lesson has helped them understand about Mama Njeri's situation (e.g., 'Selling out and having leftovers are mutually exclusive — she can add those probabilities'). On the ORANGE note they write one new question today's lesson raised for them (e.g., 'What if two events are NOT mutually exclusive — how do we add then?'). Learners post their sticky notes on the DQB in designated columns. The teacher reads aloud 3–4 green and 3–4 orange notes, clustering related questions and previewing: 'Your orange questions are EXACTLY what Lesson 5 will tackle.'	Give learners 3 minutes of silent writing time for their sticky notes — say: 'Connect your green note directly to Mama Njeri. How does mutually exclusive events help HER, not just help you pass an exam?' After posting, read selected notes aloud using a warm, affirming tone. Point to orange questions and say: 'Notice how our understanding is growing — last week we didn't even know what mutually exclusive meant, and now we're already asking about what happens when events CAN overlap. That's mathematical thinking.' Circle 2–3 orange questions on the DQB with a marker and label them 'Next investigation.'	Driving Question Board Protocol (NGSS Practice 1 — Asking Questions): The DQB externalises the class's collective growing understanding and makes the storyline progression visible. Separating 'what I now understand' (green) from 'new questions' (orange) models the scientific practice of distinguishing current evidence from remaining uncertainty — mirroring Mama Njeri's own notebook, which shows what she knows and what she still cannot predict.	Observable indicator: Green sticky notes are assessed for specificity — a note that says 'I learned probability' is insufficiently connected to the phenomenon; a note that says 'Mama Njeri can add P(sold out) + P(half unsold) = 0.85 to know there is an 85% chance of one of those outcomes' shows deep connection. Teacher collects and photographs the DQB after class for planning Lesson 5.
Model Building Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Probability of Independent Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve the cumulative probability model for Mama Njeri's stall that they began building in Lesson 1 and have revised each lesson (a hand-drawn or printed diagram showing the sample space of daily sales outcomes and associated probabilities). Today's revision task: (1) Re-label or annotate the model to mark which pairs of outcomes in Mama Njeri's sample space are mutually exclusive and which are not. (2) Add a 'Rules Panel' to the model with the formula $P(A \cup B) = P(A) + P(B)$ [for mutually exclusive events], illustrated with one worked example from Mama Njeri's data. (3) Write a 'Recommendation Box' at the	Distribute learners' model portfolios (or direct them to their exercise book model pages). Say: 'Your model is a living document — it grows every lesson as we gather more evidence. Today, it needs to show mutually exclusive events.' Circulate and prompt: 'I see you've labelled sold-out and half-unsold as mutually exclusive — can you show me WHERE on your Venn diagram section that the intersection is empty?' For the Recommendation Box, push with: 'Don't write vague advice. Write a specific probability calculation Mama Njeri can use tomorrow morning.' After peer review, cold-call 2 learners to read their Recommendation Box aloud.	Cumulative Model Revision (NGSS Practice 2 — Developing and Using Models; NGSS Practice 8 — Obtaining, Evaluating, and Communicating Information): Returning to and upgrading the same model across nine lessons builds coherence and makes knowledge growth visible to learners. The Recommendation Box forces application — learners must translate abstract probability rules into actionable advice for a real person, reinforcing the unit's core purpose: mathematics as a decision-making tool.	Observable indicator: Teacher checks the updated model for three elements — (1) correct annotation of mutually exclusive vs. non-mutually exclusive pairs, (2) accurate formula in the Rules Panel with a correct worked example using Mama Njeri's data, (3) a specific, quantitative recommendation in the Recommendation Box (not vague). Models that lack the Recommendation Box or give non-quantitative advice are flagged for follow-up at the start of Lesson 5.

	bottom of the model: 'Based on today's lesson, one thing Mama Njeri can now calculate that she couldn't before is…' Learners share their updated models with a partner for a 2-minute peer review using the prompt: 'Is every mutually exclusive pair correctly identified? Is the formula applied accurately?'	Close with: 'Every time we update this model, Mama Njeri gets closer to making a smarter decision. What's still missing from our model? Hold that thought for next lesson.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners successfully distinguish mutually exclusive from non-mutually exclusive events during the card sort, or did the majority require correction in Phase 3? If so, which event pairs caused the most confusion — and how can you address this at the start of Lesson 5? (2) Were learners able to connect the addition rule back to Mama Njeri's situation, or did they treat it as an abstract formula? Review the Recommendation Boxes in their models — specific quantitative recommendations indicate deep connection; vague advice indicates surface-level learning. (3) Did the tiered practice problems provide enough differentiation? Were higher-attaining learners genuinely challenged by the Mombasa weather problem, or did they finish early with idle time? Plan an extension task for Lesson 5. (4) Review the orange sticky notes on the DQB: do learners' new questions naturally point toward non-mutually exclusive events and the general addition rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$? If not, how will you bridge the gap in Lesson 5's opening? (5) Note: the KICD definition provided in the curriculum content contains an error — 'Mutually Exclusive Events: Events where one affects the other, like drawing cards without replacement' is actually the definition of dependent events. Ensure your teaching uses the correct definition: mutually exclusive events cannot both occur simultaneously, and $P(A \cap B) = 0$.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners sorted event-pair cards from Mama Njeri's market and Kenyan daily life into 'can happen at the same time' vs. 'cannot happen at the same time,' then physically placed outcome cards onto large Venn diagram templates to see which pairs produce an empty intersection.
What did I learn?	Two events are mutually exclusive if they cannot both occur in the same trial — their intersection is empty ($A \cap B = \varnothing$, $P(A \cap B) = 0$). For mutually exclusive events, $P(A \cup B) = P(A) + P(B)$. Mama Njeri's 'sold out' and 'half unsold' outcomes are mutually exclusive, so their probabilities can be added directly.
How does this explain the phenomenon?	Learners formalised the addition rule through a guided class discussion anchored in Mama Njeri's sales data, practised with tiered problems from Kenyan market, sports, and weather contexts, and updated their cumulative probability models with a specific quantitative recommendation for Mama Njeri's daily stocking decision.

LESSON 5 (80 minutes): Independent Events: Does One Outcome Affect the Next?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To develop learners' understanding of independent events by investigating whether the outcome of one trial affects the probability of another, and to apply the multiplication rule $P(A \cap B) = P(A) \times P(B)$ to real-life Kenyan contexts.
Knowledge	Learners will know that: (1) two events A and B are independent if the occurrence of one does not affect the probability of the other; (2) the formal test for independence is $P(A \cap B) = P(A) \times P(B)$; (3) independent events are distinct from mutually exclusive events — mutually exclusive events cannot both occur, whereas independent events can both occur but do not influence each other.
Skills	Learners will be able to: (1) determine whether two events are independent using experimental data and the multiplication rule; (2) calculate $P(A \cap B)$ for independent events; (3) distinguish between independent and mutually exclusive events with correct mathematical justification; (4) apply the concept of independence to make predictions in real-life scenarios such as Mama Njeri's mandazi sales and Eldoret seed germination.
Attitudes	Learners will appreciate that mathematical reasoning about independence helps communities — farmers in Eldoret, market traders like Mama Njeri — make evidence-based decisions rather than relying on superstition or guesswork. They will develop intellectual honesty by testing their intuitions against data.
Key Inquiry Question	How do we determine the probability of an event? How do we use probability to make decisions in real-life situations?
Purpose in Storyline	In Lessons 1–4 learners established sample spaces, calculated basic probabilities, and explored relative frequency. Now, in Lesson 5, they tackle a crucial real-world question: if Mama Njeri had a 'good day' yesterday, does that make today more or less likely to be a good day too? By formally defining and testing independence — using coin-toss experiments and the Eldoret seed germination scenario — learners gain the conceptual tool needed to model sequences of events and move closer to advising Mama Njeri on optimal daily production.
Safety Notes	No chemical or physical hazards. Ensure coins used in the experiment are handled respectfully and returned. If using printed data cards, ensure no sharp edges. Remind learners to handle Mama Njeri's notebook data (printed handout) carefully as a shared classroom resource.

B. LESSON OVERVIEW

Learners open by revisiting Mama Njeri's sales notebook data from prior lessons and debating whether yesterday's 'good day' makes today's outcome more likely to be good — surfacing the intuitive misconception that sequential events influence each other. In the Predict phase, learners commit individual written predictions. In the Observe phase, they run a structured coin-toss experiment (20 trials of two simultaneous coins) and analyse printed Eldoret seed germination data cards, recording joint frequencies. In the Explain phase, learners are introduced to the formal definition of independence — $P(A \cap B) = P(A) \times P(B)$ — and test it against their experimental results. A critical whole-class discussion explicitly confronts the misconception that independent events and mutually exclusive events are the same. In the DQB phase, learners update the Driving Question Board with new evidence and lingering questions. In the Model Building phase, learners revise their personal probability models for Mama Njeri's sales sequence, now incorporating independence as an analytical lens.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Mutually Exclusive Events Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Probability of Independent Events (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners individually read a short vignette: 'Yesterday was a great day — Mama Njeri sold 95% of her mandazi. Based on this, do you think today is more likely, less likely, or equally likely to be a good day? Write your prediction and your reasoning in your exercise book. Then share with your shoulder partner.' After partner discussion, learners write a second, revised prediction. The class hears 4–5 predictions read aloud before the teacher records two competing class positions on the board: (A) 'Yesterday's result DOES affect today' vs. (B) 'Yesterday's result does NOT affect today.'	1. Display the vignette on the board and read it aloud. Ask: 'Before we do any maths — just using your gut — write down what you think and WHY. You have 2 minutes. Silence please.' [WAIT 2 minutes.] 2. Say: 'Now share your prediction with your shoulder partner. You each have 45 seconds.' [WAIT 90 seconds.] 3. Cold-call EXACTLY 5 learners — aim for gender balance and include at least one learner who typically does not volunteer. Ask each: 'What did you predict, and what was your reasoning?' [WAIT 10 seconds after each response before commenting.] 4. Tally responses on the board under (A) and (B). Say: 'Hold your prediction — we are going to collect real evidence today to test both sides.' 5. Connect to prior lessons: 'In Lesson 4 we saw that relative frequency stabilises over many trials for Mama Njeri. Today we ask: does one trial change the probability of the next?'	Predict-Press-Commit (PPC): Learners make an individual written prediction, discuss with a partner, then commit to a class-level position. This surfaces prior knowledge and misconceptions — particularly the 'gambler's fallacy' (believing a run of good days makes a bad day 'due') — giving the teacher a live diagnostic before instruction begins. Recording two competing positions on the board creates productive cognitive tension that the lesson then resolves with evidence.	Observable indicator: Every learner has a written prediction with reasoning in their exercise book before partner talk begins. The teacher circulates and checks for the presence of reasoning (not just a yes/no). Red flag: learners who write 'it must be a bad day because she was lucky yesterday' — this signals the gambler's fallacy and should be noted for targeted questioning during the Explain phase.
Observe Phase 📹 **VIDEO:** [Mutually Exclusive Events Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Probability of Independent Events (Flexbook)] Source: CK-12 🔍 Search ARES for similar	Activity A — Coin Toss Experiment (Groups of 4): Each group receives 2 coins and a printed recording sheet. They toss BOTH coins simultaneously 40 times, recording each outcome (HH, HT, TH, TT) in a tally table. They compute the observed relative frequency of each outcome and the observed frequency of 'both heads' (HH). They also separately record how often Coin 1 landed heads and how often Coin 2 landed heads. Activity B — Eldoret Seed Germination Data Cards: Each group receives a printed card showing germination records for	1. Distribute materials. Say: 'In your groups, one person tosses, one records, one checks, one manages the tally. Rotate roles every 10 tosses.' [WAIT for groups to self-organise — 1 minute.] 2. Circulate during the coin toss. Prompt struggling groups: 'Show me your tally for HH. Now count your total tosses. What fraction is that?' [WAIT 10 seconds for response.] 3. After 40 tosses, say: 'Pause. Before you calculate anything, predict: what fraction of tosses do you expect to be HH? Write it down.' [WAIT 30 seconds.] 4. For Activity B, say: 'Read the	Parallel Data Triangulation: Learners gather evidence from two different real-world contexts — coin tosses (a pure-chance scenario) and seed germination (an agricultural scenario relevant to Kenyan farmers in Eldoret). Comparing patterns across both contexts helps learners see that the multiplicative relationship is not a coincidence of one dataset but a consistent mathematical structure. The shared class data table on the board aggregates individual group results, increasing sample size and reinforcing the Law of Large Numbers established in Lesson 4.	Observable indicators: (1) Each group's recording sheet shows 40 tally marks with correct subtotals — teacher spot-checks at least 3 groups during the experiment. (2) Learners can correctly state P(G1), P(G2), and P(G1 AND G2) from the data card without prompting. Red flag: Groups whose P(HH) is drastically different from 0.25 — teacher asks them to recount their tallies and checks for recording errors. Green flag: A group that spontaneously says 'P(HH) is close to P(H on Coin 1) times P(H on Coin 2)' — affirm and ask them to explain to

readings	200 maize seeds planted in Eldoret. For each seed, the card shows whether Seed 1 germinated (G1) and whether the adjacent Seed 2 germinated (G2). Learners tally: P(G1), P(G2), P(G1 and G2), then compare P(G1 and G2) to P(G1) × P(G2). Both activities feed into a shared class data table built on the board.	data card carefully. Your job is to find three numbers: P(G1), P(G2), and P(G1 AND G2). Record them in your table.' 5. Cold-call 3 groups to share their P(G1 AND G2) values. Record all on the board. Ask: 'Are these values similar across groups? Why might they differ slightly?' [WAIT 15 seconds.] 6. Ask: 'What do you notice when you compare P(G1 AND G2) with P(G1) × P(G2)?' Cold-call 2 learners. [WAIT 12 seconds after each.]		their group.
Explain Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Probability of Independent Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher formally introduces the definition of independent events and the multiplication rule. Learners then work through three structured examples in their exercise books: (1) Coin toss — verifying P(HH) = P(H) × P(H) = ½ × ½ = ¼ against their experimental data; (2) Mama Njeri's sales — using the class-compiled relative frequencies from Lesson 4 to test whether 'good day on Monday' and 'good day on Tuesday' are independent; (3) Mutually exclusive vs. independent — a side-by-side worked example: 'Event A = Mama Njeri sells out. Event B = Mama Njeri sells nothing. Are these mutually exclusive? Are they independent? Justify using the definitions.' Learners write formal definitions of both terms in their own words, then swap books with a partner to peer-assess the definitions against the board criteria.	1. Say: 'Based on your data, I want to propose a definition. Listen carefully.' Write on board: 'Two events A and B are INDEPENDENT if and only if P(A ∩ B) = P(A) × P(B).' Read it aloud twice. Ask: 'In your own words, what does this mean? Write one sentence.' [WAIT 90 seconds.] 2. Cold-call 4 learners. After each, ask the class: 'Thumbs up if you agree with that wording — thumbs sideways if you'd add something.' [WAIT 10 seconds per response.] 3. Work through Coin Toss example on the board step-by-step, narrating each line. Ask: 'Does our experimental value of P(HH) match the theoretical value? What does any small difference tell us?' [WAIT 12 seconds.] 4. For Mama Njeri example, say: 'Get out Lesson 4's relative frequency table. Find P(good day) from the data. Now — if days are independent — what should P(good Monday AND good Tuesday) equal? Calculate it.' [WAIT 3 minutes for individual work.] Cold-call 2 learners to share calculations. 5. CRITICAL MISCONCEPTION MOMENT — Say: 'I have heard some people	Claim–Evidence–Reasoning (CER) with Misconception Deconstruction: Learners make a Claim (two events are/are not independent), provide Evidence (their calculated values of P(A ∩ B) and P(A) × P(B)), and Reasoning (whether these match, and what that means). The explicit side-by-side comparison of 'independent' and 'mutually exclusive' uses a Conceptual Contrast strategy — placing the correct understanding directly alongside the most common error — so learners build correct schema rather than patching misconceptions later.	Observable indicators: (1) In the Mama Njeri calculation, learners correctly multiply two relative frequencies to get P(good Monday AND good Tuesday) — teacher scans at least 6 books during the 3-minute work period. (2) During peer-assessment, learners can identify at least one specific error or gap in their partner's definition. Red flag: A learner who writes 'independent means they cannot happen together' — this is the mutually exclusive definition; teacher addresses individually. Exit check: Teacher asks 3 cold-called learners to complete the sentence: 'Events A and B are independent if __________.' Correct responses must reference the multiplication rule.

		say that independent events and mutually exclusive events are the same thing. Let's destroy that misconception right now.' Write both definitions side-by-side. Ask: 'Can two events be BOTH independent AND mutually exclusive at the same time? Discuss with your partner for 60 seconds.' [WAIT 60 seconds.] Cold-call 3 learners. Guide class to the conclusion: if $P(A) > 0$ and $P(B) > 0$ and A and B are mutually exclusive, then $P(A \cap B) = 0$, which does NOT equal $P(A) \times P(B)$ — so they cannot be both. 6. Learners write formal definitions and peer-assess. Circulate and stamp or tick books where definitions are accurate.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Mutually Exclusive Events Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Probability of Independent Events (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives one sticky note (or a small card if sticky notes are unavailable). They write ONE of the following — their choice: (a) a piece of new evidence gathered today that helps answer the driving question, (b) a question that today's lesson raised for them, or (c) a connection between today's concept and Mama Njeri's situation. Learners post their cards on the class Driving Question Board under the column: 'Lesson 5 — Independence.' The teacher reads 5–6 cards aloud, grouping them into 'Evidence we now have' and 'Questions still open.' The class briefly discusses: 'How does knowing whether daily sales are independent change what advice we can give Mama Njeri?'	1. Say: 'Take your sticky note. Think about everything we did today — the coins, the seeds, the definitions. Write ONE thing: new evidence, a new question, or a connection to Mama Njeri. You have 2 minutes.' [WAIT 2 minutes. Silence.] 2. Invite learners to post cards on the board. Give 90 seconds for posting. 3. Read 5–6 cards aloud, think-aloud while sorting: 'This one gives us evidence — I'll put it here. This one is still a question — let's keep it visible.' 4. Ask: 'If Mama Njeri's daily sales ARE independent — meaning yesterday does not affect today — how does that change how she should plan? Discuss in pairs for 60 seconds.' [WAIT 60 seconds.] Cold-call 2 pairs to share. 5. Say: 'We now have strong evidence that independent events follow the multiplication rule. But we still have open questions — for example, are ALL of Mama Njeri's days truly	Driving Question Board Curation: The DQB is a living epistemic artefact — a public record of the class's evolving understanding. By asking learners to choose between posting evidence, a question, or a connection, the teacher differentiates the cognitive demand and ensures all learners contribute meaningfully regardless of confidence level. Grouping cards into 'evidence' and 'open questions' models the scientific practice of distinguishing what is known from what remains to be investigated.	Observable indicators: (1) Every learner posts a card — teacher scans the board to confirm participation. (2) Cards under 'Evidence' reference today's mathematical content (multiplication rule, independence, experimental data) rather than vague statements. (3) At least one card raises a genuinely novel question (e.g., 'What if events are NOT independent — how do we calculate then?'). Red flag: Cards that conflate independence with mutual exclusivity — teacher flags these anonymously in the whole-class discussion as 'a question worth clearing up.'

		independent? We will keep investigating.'		
Model Building Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Probability of Independent Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their personal probability model of Mama Njeri's mandazi sales — a diagram or written model they have been building since Lesson 1. Today they must revise or annotate the model to show: (1) that consecutive daily sales events appear to be independent (based on today's evidence); (2) the calculation P(good Monday AND good Tuesday) = P(good day)² using the relative frequency from Lesson 4; (3) a clear label distinguishing 'independent' from 'mutually exclusive' within the model. Learners also write a 2-sentence 'advice note' to Mama Njeri: 'Based on today's evidence, here is what independence means for your planning...' Groups share models with an adjacent group for a 2-minute gallery walk and leave one written comment on a sticky note.	1. Say: 'Open your models from Lesson 1. You are now going to upgrade them with today's evidence. You have 8 minutes — work individually.' [WAIT 8 minutes. Circulate actively, prompting with: 'Where does independence appear in your model?' and 'Have you included the calculation?'] 2. After 8 minutes, say: 'Write your advice note to Mama Njeri. Two sentences. Use the word INDEPENDENT.' [WAIT 3 minutes.] 3. Announce the gallery walk: 'Stand up, move to the group on your left. Read their model and advice note. Leave one written comment — either a question or an agreement. You have 2 minutes.' [WAIT 2 minutes.] 4. Cold-call 3 learners to read their advice note to the class. After each, ask: 'Does this advice correctly use what we learned today? Thumbs up or sideways.' [WAIT 10 seconds per response.] 5. Close: 'Today we proved — with coins, with seeds, and with Mama Njeri's data — that independent events follow the multiplication rule. Next lesson, we will ask: what happens when events are NOT independent? How does that change everything?'	Progressive Model Revision (PMR): Learners do not build a new model each lesson — they revise a single cumulative model, making their growing understanding visible as annotations, corrections, and additions. This mirrors how scientists refine models as new evidence arrives. The 'advice note' to Mama Njeri is a transfer task — it requires learners to apply abstract mathematical reasoning to the original anchoring phenomenon in plain language, consolidating conceptual understanding.	Observable indicators: (1) Revised model explicitly includes the label 'independent events' and a correct multiplication calculation using the Lesson 4 relative frequency. (2) Advice note uses the word 'independent' correctly in context — e.g., 'Because each day's sales are independent, knowing yesterday was good does not change the probability of today being good.' Red flag: A model that shows an arrow from 'Day 1 outcome' to 'Day 2 outcome' suggesting influence — teacher asks the learner: 'What does your arrow mean? Does today's result change tomorrow's probability? What did our data show?' Green flag: A model that correctly computes P(two consecutive good days) and labels it with the multiplication rule.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) MISCONCEPTION CHECK — How many learners still conflated independence with mutual exclusivity during the Explain phase? Did the side-by-side deconstruction activity resolve this for most, or do I need a targeted re-teach at the start of Lesson 6? (2) PARTICIPATION — Did the cold-call strategy reach learners who do not typically volunteer? Were gender and ability groups balanced across the five cold-call moments? (3) EVIDENCE QUALITY — Were the DQB cards specific and mathematically grounded, or were they vague? If vague, consider modelling a stronger example card at the start of Lesson 6. (4) MODEL PROGRESSION — Are learners' cumulative models becoming more sophisticated with each lesson, or are some learners still drawing the same basic diagram from Lesson 1? Identify 3 learners whose models need individual feedback before Lesson 6. (5) PHENOMENON CONNECTION — Did every phase visibly connect back to Mama Njeri's sales? If any phase felt disconnected, note how to re-anchor it next time. (6) TIMING — The coin-toss experiment can run over time if groups lose focus during tallying. Consider pre-assigning roles before distributing coins in future iterations.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students tossed two coins simultaneously 40 times per group and recorded outcomes (HH, HT, TH, TT), then analysed printed Eldoret seed germination data cards showing whether paired seeds germinated. They tallied joint frequencies and computed observed relative frequencies for each outcome across both activities.
What did I learn?	Students learned that two events are independent if the occurrence of one does not affect the probability of the other, formally defined as $P(A \cap B) = P(A) \times P(B)$. They verified this rule against their coin-toss and seed germination data. They also learned that independent events and mutually exclusive events are distinct concepts — mutually exclusive events cannot both occur, while independent events can both occur but do not influence each other.
How does this explain the phenomenon?	Students explained why Mama Njeri's daily mandazi sales appear to behave as independent events — yesterday's outcome does not change the probability of today's outcome. Using the multiplication rule and relative frequency data from Lesson 4, they calculated the probability of two consecutive good sales days, and revised their cumulative probability models to reflect this understanding. They wrote a two-sentence evidence-based advice note to Mama Njeri applying the concept of independence to her daily production planning.

LESSON 6 (80 minutes): The Addition Law of Probability — Overlapping Events in Mama Njeri's World	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To extend students' understanding of probability from mutually exclusive events to overlapping events, enabling them to calculate $P(A \cup B)$ using the full Addition Law of Probability.
Knowledge	Students will know that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, that mutually exclusive events are a special case where $P(A \cap B) = 0$, and that Venn diagrams model the relationship between overlapping event sets within a sample space.
Skills	Students will be able to identify whether two events are mutually exclusive or overlapping, draw and interpret two-circle Venn diagrams for combined events, apply the Addition Law to calculate probabilities of unions of events, and connect calculated probabilities to real-world decision-making contexts such as Mama Njeri's sales and Nakuru rainfall data.
Attitudes	Students will appreciate that mathematical tools can make sense of complex, overlapping real-world uncertainties, and will develop confidence in using formal probability laws to reason about everyday decisions.
Key Inquiry Question	How do we determine the probability of an event? How do we use probability to make decisions in real-life situations?
Purpose in Storyline	In Lesson 5, students established the Complementary Rule and explored mutually exclusive events — situations where two events cannot both happen on the same day (e.g., Mama Njeri either sells out OR she doesn't). Now, in Lesson 6, the storyline advances to a more realistic and messier question: what about events that CAN overlap? On a given day at Gikomba Market, Mama Njeri might experience BOTH high sales AND good weather — these are not mutually exclusive. Students discover that simply adding $P(A) + P(B)$ double-counts the overlap, and the Addition Law corrects this. By the end of the lesson, students can estimate: 'On how many days might Mama Njeri experience high sales OR good weather (or both)?' — a combined probability that directly informs her morning decision about how many mandazi to fry.
Safety Notes	No physical hazards anticipated. Ensure inclusive classroom participation — some students may feel anxious about Venn diagram notation. Normalise errors as part of the sensemaking process. If using printed Mama Njeri data cards, ensure equitable distribution across groups.

B. LESSON OVERVIEW

Lesson 6 is the conceptual centrepiece of the Addition Law sequence. Students begin by revisiting the Driving Question Board and their accumulated evidence about Mama Njeri's mandazi sales patterns. They make a prediction about a combined-event problem (high sales OR good weather) before discovering, through a structured Venn diagram activity using Nakuru rainfall and sales data, that naively adding two probabilities can overcount outcomes. Through guided sensemaking, students derive and apply $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, recognise mutually exclusive events as a special case, and update both their Venn diagram models and their DQB entries. The lesson closes with students explicitly connecting the Addition Law back to advising Mama Njeri — how many mandazi should she fry on a day when either high sales OR bad weather (which reduces foot traffic) could occur?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Venn Diagrams Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Addition of Fractions: Walking to School (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students revisit the DQB from Lessons 1–5 and respond individually in their notebooks to a new prediction prompt: 'Mama Njeri's records show that on 40 out of 100 days she had HIGH sales, and on 35 out of 100 days there was HEAVY RAIN at Gikomba Market. On how many of those 100 days do you think she had HIGH SALES OR HEAVY RAIN? Write your best guess and explain your reasoning in 2–3 sentences.' Students share predictions with a shoulder partner, then a few are cold-called to share with the class. Teacher records all predicted numbers on the board without evaluating them.	Begin by pointing to the DQB: 'Look at what we have built together over the last five lessons. We know how to calculate simple probabilities, use the complementary rule, and handle mutually exclusive events. Today we meet a new challenge — what happens when two events CAN happen on the same day?' Distribute the prediction prompt card or write it on the board. Say: 'Read the prompt silently. Write your prediction and reasoning — you have 3 minutes. There is no wrong answer right now.' After 3 minutes, say: 'Turn to your shoulder partner. Share your number and your reason — 90 seconds.' Allow pair talk, then cold-call exactly 4 students (aim for a range of predicted values — e.g., one who says 75, one who says 40, one who says 35, one who says something else). For each response, ask: 'What reasoning led you to that number?' Record all values on the board in a visible 'Our Predictions' column. Do NOT confirm or correct yet. Close with: 'Hold onto your prediction — by the end of this lesson, you will know who was closest and WHY the mathematics works the way it does.'	Elicit-and-Record Prediction Strategy — by surfacing a range of student guesses (many will instinctively add 40 + 35 = 75, others may guess lower), the teacher creates a productive cognitive conflict that the lesson resolves. Recording predictions publicly creates accountability and a reference point for the 'Explain' phase reveal.	Observable indicator: Can students articulate a reasoning strategy (even an incorrect one) for combining two event counts? Listen for students who say 'I just added them' (revealing the double-counting misconception to be addressed) versus students who intuit overlap ('some rainy days might also be high-sales days'). Note these students — the second group can be leveraged as peer explainers in Phase 3.
Observe Phase 📹 **VIDEO:** Venn Diagrams Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Addition of Fractions:	Groups of 4 receive the 'Nakuru Weather & Sales Data Card' — a table of 20 simulated days showing, for each day: whether it was a HIGH SALES day (H) and whether there was HEAVY RAIN (R). (Sample data is designed so that 8 days are H, 7 days are R, and 3 days are BOTH H and R, giving 12 unique days that are H or R.) Students first sort the 20 day-cards (small printed slips, or	Distribute data cards and blank A3 paper. Say: 'Your job right now is NOT to calculate probability formulas — your job is to be detectives. Sort the days, draw the Venn diagram, and fill in the counts. You have 12 minutes.' Circulate actively. At the 4-minute mark, prompt any group that has not started the Venn diagram: 'Where do the days that are BOTH H and R go on your diagram?	Concrete-to-Abstract Venn Diagram Sorting — students physically categorise data before abstracting to notation. This grounds the symbolic formula in observable counting, reducing the risk of rote application. The act of placing the 3 'overlap' days in the intersection region makes the double-counting error viscerally visible: students can point to it on their own diagram.	Observable indicator: Does each group correctly place days that are BOTH H and R in the intersection (not in both outer circles)? Does the group's $n(H \cup R)$ come from counting all filled regions (not from adding $n(H) + n(R)$)? Watch for groups that write 15 in the union — this is the key double-counting error and should be surfaced publicly in Phase 3.

Walking to School (PLIX) Source: CK-12 🔍 Search ARES for similar readings	circled entries on a table) into four categories: H only, R only, Both H and R, and Neither. They then draw a two-circle Venn diagram on a large sheet of paper, labelling Circle A = HIGH SALES and Circle B = HEAVY RAIN, and placing the correct counts in each region. Groups record: $n(H) = ?$, $n(R) = ?$, $n(H \cap R) = ?$, $n(H \cup R) = ?$ by physically counting regions of their diagram. They also compute the fractions out of 20 for each region.	Show me with your finger.' At the 8-minute mark, pose a focusing question to the whole class: 'When you count $H \cup R$ — meaning HIGH SALES or HEAVY RAIN or BOTH — are you counting the middle section once or twice if you just add $n(H) + n(R)$?' Allow 10 seconds WAIT TIME before cold-calling 2 groups to respond. After groups finish, select 2 groups with correctly completed diagrams to display their work under the visualiser or on the board. Ask the class: 'What do you notice about the number in the overlapping region?' Ensure all groups have: $n(H) = 8$, $n(R) = 7$, $n(H \cap R) = 3$, $n(H \cup R) = 12$.		
Explain Phase 📹 **VIDEO:** Venn Diagrams Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Addition of Fractions: Walking to School (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher facilitates a whole-class derivation of the Addition Law, anchored in the Venn diagrams groups just built. Students follow along in their notebooks, then work through three tiered practice problems — first with the Nakuru data, then with a Kisumu fishing cooperative scenario, then with an abstract problem — before returning to the original prediction prompt to see whose estimate was closest and why. Students write a 'Law Statement' in their own words in their notebooks and annotate their Venn diagram with the formula.	Begin the explanation by returning to the prediction board: 'Many of us predicted 75. Let's see what our Venn diagrams actually show.' Point to a displayed diagram. 'How many days total are in Circle H?' [8] 'How many in Circle R?' [7] 'If I add those, I get 15. But your diagram shows only 12 days shaded. Why?' Allow 10 seconds WAIT TIME. Cold-call 3 students. Guide the class to articulate: 'The 3 days in the middle were counted TWICE — once in H and once in R — so we must subtract them once.' Write on the board: $n(H \cup R) = n(H) + n(R) - n(H \cap R) = 8 + 7 - 3 = 12$. Say: 'Now divide everything by 20 to get probability.' Write: $P(H \cup R) = P(H) + P(R) - P(H \cap R) = 8/20 + 7/20 - 3/20 = 12/20 = 0.6$. Box the formula. Say: 'This is the Addition Law of Probability. Write it in your notebook and circle it.' Pause 30 seconds for students to write. Then: 'Now — when are $P(A \cap B) = 0$? Turn and talk — 60 seconds.'	Anchor-Formula-Apply Sequence — the teacher anchors the formula derivation in the students' own Venn diagram data (not a textbook example), making the subtraction step feel necessary rather than arbitrary. The three-tiered practice (familiar data → new Kenyan context → return to phenomenon) scaffolds transfer while continuously reinforcing the anchoring phenomenon.	Observable indicators: (1) In Practice Problem 1, students should write $P(H \cup R) = 8/20 + 7/20 - 3/20 = 12/20$ — check for the subtraction step being present. (2) In Practice Problem 2, students should get $P = 0.5 + 0.4 - 0.2 = 0.7$ — listen for pairs who forget to subtract. (3) In the Mama Njeri problem, students should get $0.40 + 0.35 - 0.15 = 0.60$, not 0.75 — the 0.75 answer is the diagnostic for persisting double-counting misconception. Collect notebook work from 6 randomly selected students at the end of this phase.

		Cold-call 2 students. Confirm: mutually exclusive events have no overlap, so the formula reduces to what we learned in Lesson 5. Then pose Practice Problem 1: 'Using the Nakuru data, what is the probability that a randomly chosen day is a HIGH SALES day or a HEAVY RAIN day? Use the formula.' Students work individually — 3 minutes. Then Practice Problem 2 (Kisumu): 'In a Kisumu fishing cooperative, P(large catch) = 0.5, P(calm weather) = 0.4, P(large catch AND calm weather) = 0.2. Find P(large catch OR calm weather).' Students work in pairs — 4 minutes. Cold-call 2 pairs. Finally, return to the Mama Njeri prediction: 'Now recalculate — if P(high sales) = 0.40, P(heavy rain) = 0.35, and P(both) = 0.15, what is P(high sales OR heavy rain)?' Students work individually — 3 minutes. Reveal answer: 0.60. Ask: 'Whose prediction was closest? What did they understand intuitively that others missed?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Venn Diagrams Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Addition of Fractions: Walking to School (PLIX) Source: CK-12 🔍 Search ARES for similar	Each student receives a sticky note in a new colour (designated for Lesson 6). They respond to two DQB prompts: (1) 'What new evidence does the Addition Law give us to help answer: How can mathematics help Mama Njeri make better decisions?' and (2) 'Write one new question this lesson raises for you — something you are still wondering about.' Students post sticky notes in the appropriate columns of the DQB ('New Evidence' and 'New Questions'). The class does a 3-minute gallery walk to read peers' contributions, then two students are cold-called to share one insight	Say: 'The DQB is our running record of what we know and what we still wonder. Today we added a powerful new tool. Take your Lesson 6 sticky note — the green one. On side one, write how the Addition Law helps us answer our driving question about Mama Njeri. Be specific — mention numbers or the formula. On side two, write a genuine question this lesson raised.' Allow 4 minutes of individual writing. Then: 'Walk to the board, post your note, and spend 3 minutes reading at least 4 other notes.' After the gallery walk, cold-call 2 students: 'Tell me one idea from someone else's note that	DQB Metacognitive Anchoring — the dual-prompt sticky note (evidence + new question) requires students to simultaneously consolidate learning and identify the boundary of their current understanding. Public posting and gallery walk create a shared intellectual record that visually shows the class's cumulative progress toward answering the driving question.	Observable indicator: Do students' 'New Evidence' notes reference the Addition Law formula or the concept of overlapping events specifically, rather than giving a vague response? Do 'New Questions' reveal genuine conceptual curiosity (e.g., 'Does rain actually cause fewer sales, or is the overlap just a coincidence?') or surface misconceptions to address in Lesson 7? Photograph the DQB board at the end of this phase for teacher records.

readings	and one question from the board.	you found interesting.' Then point to any 'New Questions' sticky notes that preview Lesson 7 content (conditional probability, or 'does weather CHANGE the probability of high sales?') and say: 'These questions are exactly where we are heading next. Hold onto them.'		
Model Building Phase 📹 **VIDEO:** Venn Diagrams Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Addition of Fractions: Walking to School (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their cumulative probability model of Mama Njeri's situation — a Venn diagram / probability map begun in earlier lessons. They now add a new layer: a two-circle Venn diagram for HIGH SALES (A) and GOOD WEATHER (B), populated with the lesson's values ($P(A) = 0.40$, $P(B) = 0.35$, $P(A \cap B) = 0.15$, $P(A \cup B) = 0.60$). They annotate the diagram with the Addition Law formula and write a one-sentence 'Decision Advice' box: 'Based on this model, Mama Njeri should expect a HIGH SALES or GOOD WEATHER day approximately ___ out of every 10 days, so she should consider ___.' Students also note on their model: 'What this model still CANNOT tell us: ___.' Groups share their 'Decision Advice' sentences with a neighbouring group and give one piece of feedback.	Say: 'Every lesson, we make our model of Mama Njeri's world more complete and more honest about what it can and cannot do. Open your model from previous lessons.' Give students 2 minutes to locate their models. Then: 'Add today's Venn diagram layer. Use the values we calculated. You have 7 minutes — include the formula, the numbers in each region, and your Decision Advice sentence.' Circulate and prompt: 'What does the area OUTSIDE both circles represent? Add that to your diagram.' [Days with neither high sales nor good weather: $1 - 0.60 = 0.40$ of days.] At the 5-minute mark, prompt the 'Still CANNOT tell us' annotation: 'An honest model admits its limits. What does even this improved model not explain about why individual days vary?' After 7 minutes, say: 'Share your Decision Advice sentence with the group next to you. Give them one star — something strong — and one wish — something to improve or add.' Allow 3 minutes for peer feedback. Close: 'Next lesson, we will ask an even deeper question — if it IS raining today, does that change the probability of high sales? That is conditional probability, and it will make our model even more powerful for Mama Njeri.'	Cumulative Model Revision with Epistemic Humility Annotation — by requiring both a 'Decision Advice' (what the model can do) and a 'Still CANNOT tell us' annotation (what the model cannot do), students develop scientific and mathematical epistemic practices: models are powerful but incomplete, and honest reasoning names both. This directly mirrors how real data analysts advise businesses like Mama Njeri's.	Observable indicator: Does the student's Venn diagram correctly show four regions (A only = 0.25, B only = 0.20, $A \cap B = 0.15$, Neither = 0.40) that sum to 1.0? Does the 'Decision Advice' sentence use the probability 0.60 (or equivalent fraction) correctly? Does the 'Still CANNOT tell us' annotation reflect genuine understanding of the model's limits (e.g., 'This does not tell us whether rain causes fewer sales or if they just happen together by coincidence') rather than a trivial observation? Collect 4 models to review before Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did students arrive at the Addition Law formula through the Venn diagram evidence, or did they seem to be waiting to be told the formula? If the latter, consider extending the Phase 2 sorting activity in future iterations to give more time for groups to articulate the double-counting problem in their own words before the Phase 3 derivation. (2) How many students wrote 0.75 (the double-counting error) in the final Mama Njeri calculation? If more than 30% did, open Lesson 7 with a brief re-engagement using the Venn diagram from this lesson. (3) Were the DQB sticky notes in Phase 4 substantively connected to the formula and phenomenon, or were they generic? Consider modelling a high-quality DQB entry before students write next time. (4) Did the 'Still CANNOT tell us' annotation generate genuine epistemic conversation, or did students treat it as a formality? This annotation is a high-leverage thinking move — invest more time in it in Lesson 7 when conditional probability will reveal exactly the limitation students named today. (5) Note any student who articulated the conditional probability question spontaneously ('Does rain actually reduce sales, or is it just a coincidence?') — this student could lead the opening inquiry in Lesson 7.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students sorted Nakuru weather and sales data cards into four Venn diagram regions (HIGH SALES only, HEAVY RAIN only, BOTH, NEITHER) and counted $n(H \cup R) = 12$ out of 20 days, observing that simply adding $n(H) + n(R) = 15$ overcounts by exactly the 3 days in the intersection.
What did I learn?	Students learned the Addition Law of Probability: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, recognised that mutually exclusive events are the special case where $P(A \cap B) = 0$ (reducing the law to the rule from Lesson 5), and applied the law to calculate that on approximately 60% of days, Mama Njeri can expect HIGH SALES or HEAVY RAIN (or both) — directly informing her daily mandazi production decision.
How does this explain the phenomenon?	The double-counting problem explains why we must subtract $P(A \cap B)$: days that belong to BOTH events are included once in $P(A)$ and once in $P(B)$, so adding them directly inflates the probability. The Venn diagram makes this visible — the intersection region is counted in both circles, and subtracting it once gives the true count of days in the union. This is why Mama Njeri cannot simply add separate probabilities when events overlap in her real sales world.

LESSON 7 (80 minutes): Multiplication Law of Probability: Independent and Dependent Events	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students derive and apply the multiplication law of probability for both independent and dependent events, connecting multi-step probabilistic reasoning to real-world Kenyan decision-making contexts.
Knowledge	Students will know that: (1) For independent events A and B, $P(A \cap B) = P(A) \times P(B)$; (2) For dependent events A and B, $P(A \cap B) = P(A) \times P(B\|A)$, where $P(B\|A)$ is the conditional probability of B given A has occurred; (3) Two events are independent if the occurrence of one does not affect the probability of the other; (4) Sampling without replacement creates dependent events because each selection changes the remaining sample space.
Skills	Students will be able to: (1) Distinguish between independent and dependent events in context; (2) Derive the multiplication law from tree diagrams and sample space reasoning; (3) Apply $P(A \cap B) = P(A) \times P(B)$ to independent event problems; (4) Apply $P(A \cap B) = P(A) \times P(B\|A)$ to dependent event problems involving sampling without replacement; (5) Interpret multiplication law results to advise real-life decision-making scenarios.
Attitudes	Students will appreciate that: (1) Precise mathematical reasoning enables better decisions under uncertainty; (2) Careful attention to whether events are independent or dependent is essential for accurate probability calculations; (3) Mathematics is a powerful tool for understanding and navigating everyday Kenyan economic and agricultural challenges.
Key Inquiry Question	How do we determine the probability of an event? How do we use probability to make decisions in real-life situations?
Purpose in Storyline	In previous lessons, students built understanding of sample spaces, single-event probability, the addition law, and complementary events using Mama Njeri's mandazi data. Now students must grapple with multi-step events — what is the probability that TWO things both happen? This lesson equips students to calculate compound probabilities by multiplying, a critical tool for Mama Njeri who must consider the joint likelihood of multiple uncertain factors (e.g., a good sales day AND a supplier delivering quality maize flour). The multiplication law is the next major piece of evidence students need to answer the driving question: how can mathematics help predict and plan for uncertain outcomes?
Safety Notes	No physical hazards in this lesson. Ensure respectful group discussion norms are maintained. If using maize seeds as physical manipulatives, ensure no students have seed allergies; seeds used for handling only, not consumption.

B. LESSON OVERVIEW

In this 80-minute lesson, students derive and apply the multiplication law of probability through a sequence of carefully scaffolded experiences anchored in the Mandazi Stall Mystery. Beginning with a prediction task about drawing maize seeds from a bag, students use tree diagrams to discover for themselves why probabilities multiply across sequential events. They then distinguish between independent events (flipping a coin twice) and dependent events (drawing maize seeds without replacement), deriving both forms of the multiplication law: $P(A \cap B) = P(A) \times P(B)$ and $P(A \cap B) = P(A) \times P(B|A)$. The lesson culminates in students advising Mama Njeri on a compound business decision using their new tool. This is Lesson 7 of 9 in the evidence-gathering sequence; students add the multiplication law as a key piece of evidence on the Driving Question Board toward answering how mathematics can help make smarter decisions under uncertainty.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Conditional Probability Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Multiplication of Fractions by Whole Numbers (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Students are shown a physical scenario (or drawn diagram on the board): a cloth bag contains 5 maize seeds — 3 that will germinate (labelled G) and 2 that will not (labelled NG). Mama Njeri's cousin, a smallholder farmer in Rift Valley, randomly picks one seed, plants it, then picks a second seed (without putting the first back) and plants it. Students individually record their prediction for the question: 'What is the probability that BOTH seeds will germinate?' Students write a numerical guess and a one-sentence justification in their exercise books before any instruction occurs.	Display the bag diagram on the board with clear labels: '3 Germinate (G), 2 Do Not Germinate (NG) — 5 seeds total. Seeds are drawn one at a time WITHOUT replacement.' Say exactly: 'Before we do any mathematics, I want your instinct. In your exercise book, write your best guess for the probability that BOTH seeds germinate, and write ONE sentence explaining your reasoning. You have 90 seconds — go.' [Allow 90 seconds of silent individual writing.] After time, say: 'Turn to your partner. Share your prediction and your reason. You have 60 seconds.' [Allow 60 seconds.] Cold-call 4 students (select a mix of confidence levels — identify these students during partner talk by circulating): 'Zawadi, what did you predict and why?' [WAIT TIME: 10–12 seconds after naming student before prompting.] Record 3–4 different student predictions on the board without affirming or correcting. Say: 'We have different predictions on the board. By the end of today, you will know exactly how to calculate this — and you will be able to check whose prediction was closest. Keep your prediction written down.'	Prediction with Justification — by requiring students to commit to a written numerical guess with reasoning BEFORE instruction, this strategy activates prior knowledge of fractions and single-event probability, creates cognitive dissonance when predictions conflict (motivating genuine inquiry), and gives the teacher real-time formative data on student starting points. Recording multiple predictions on the board without judgment establishes that all reasoning attempts are valued.	Circulate during the 90-second writing window. Observable indicators of readiness: (1) Student writes a fraction between 0 and 1 (not a percentage or word like 'likely') — indicates prior probability notation is intact; (2) Student's justification references the number of seeds — indicates they are engaging with the sample space; (3) RED FLAG — student writes 3/5 × 3/5 = 9/25 without adjustment for 'without replacement' — this is the key misconception to surface in Phase 3; note which students do this.
Observe Phase 📹 **VIDEO:** [Conditional Probability Principles] Source: CK-12 🔍 Search ARES for similar videos	Students work in groups of 4 to build a TREE DIAGRAM for the two-draw maize seed scenario. Each group receives a worksheet with a partially drawn tree diagram (first branch drawn: G and NG from the bag of 5). Students must: (1) Complete the second set of branches, carefully adjusting the denominator for each branch to	Distribute the tree diagram worksheet. Say: 'In your groups, complete the tree diagram for Mama Njeri's cousin's seed scenario. Pay very close attention to the denominators on the SECOND set of branches — they are different from the first. I am not going to tell you how yet; use your group's reasoning. You have 12	Tree Diagram Construction with Structured Comparison — tree diagrams make the sequential, multiplicative structure of compound probability visible and concrete. By having students build BOTH a dependent-event and an independent-event tree diagram side by side, the critical distinction between adjusting and not	During circulation, look for: (1) SUCCESS — second-branch denominators correctly adjusted to 4 in the seed scenario (e.g., if first draw was G, second branch shows 2G remaining from 4 seeds: P = 2/4); (2) COMMON ERROR — student keeps denominator as 5 on the second branch — this is the independent-events assumption

📖 **HTML:** [Multiplication of Fractions by Whole Numbers (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	reflect that the first seed has been removed; (2) List all possible outcomes (GG, GNG, NGG, NGNG) and their pathway probabilities; (3) Verify that all pathway probabilities sum to 1. Students then repeat the tree diagram for a SECOND scenario: flipping a 20-cent coin twice (independent events) to compare. After completing both tree diagrams, groups answer three comparison questions on the worksheet: (a) What changed in the denominator between the first and second draw in the seed scenario? (b) What stayed the same in the coin scenario? (c) Where do you see multiplication appearing in your tree diagram calculations?	minutes.' [Circulate actively — do not answer 'is this right?' questions; instead redirect: 'What happened to the total number of seeds after the first one was removed?'] At the 7-minute mark, say: 'If your first tree is complete, start the coin-flip tree diagram. Notice: does removing one coin flip change the next one?' [After 12 minutes, bring class to attention.] Display a correct completed tree diagram on the board (pre-prepared on a flipchart or projected). Say: 'Look at the pathway from G on the first branch to G on the second branch. What numbers did you multiply together to get the probability of that pathway?' [WAIT TIME: 12 seconds.] Cold-call 2 groups to share their answers to comparison questions (a), (b), and (c). Say: 'We are going to name what you just discovered. But first — look at your pathway probabilities. What operation connects P(first G) and P(second G | first was G)?' [WAIT TIME: 10 seconds.] Cold-call 3 individual students.	adjusting the denominator becomes observable, not merely told. The three comparison questions are a Structured Academic Controversy scaffold that forces groups to articulate the mathematical difference before the teacher names it. This builds conceptual understanding rather than procedural mimicry.	applied incorrectly; do NOT correct immediately; mark this group for targeted questioning during Phase 3; (3) SUCCESS on coin tree — both second branches show 1/2 regardless of first outcome — student understands independence. After groups share comparison question (c), confirm understanding if students use the word 'multiply' or show the multiplication of fractions along a pathway.
Explain Phase 📹 **VIDEO:** [Conditional Probability Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Multiplication of Fractions by Whole Numbers (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class derivation of BOTH forms of the multiplication law, using the tree diagram evidence students just produced. Students record the derivations in their exercise books with worked examples. Part A — Independent Events: Using the coin-flip tree, students formalise P(A ∩ B) = P(A) × P(B) and verify with a new example (probability that Mama Njeri has a 'good sales day' on both Monday AND Tuesday, if each day independently has P(good day) = 3/5 based on her notebook data from earlier lessons). Part B —	Begin with: 'Let's use your tree diagram evidence to build a law.' Write on the board: 'For the coin flip: P(Heads on flip 1) = 1/2. P(Heads on flip 2) = 1/2. P(HH) = ?' Point to a student's correct tree diagram answer. 'We can see from your tree that P(HH) = 1/4. And 1/2 × 1/2 = 1/4. So we write: P(A ∩ B) = P(A) × P(B) — this works when events are INDEPENDENT.' Write the formula clearly, boxed. 'Now the seeds. P(first seed germinates) = 3/5. But what is the probability the second germinates, given the first one already did?' [WAIT TIME: 12	Derivation from Evidence + Misconception Confrontation — rather than presenting the multiplication law as a given formula, the teacher derives it directly from the tree diagram evidence students just produced, making the law feel discovered rather than imposed. This builds conceptual ownership. Directly naming and examining the P(A) × P(B) misconception (treating dependent events as independent) is a Bridging Analogy strategy: students who made that error are not wrong in their reasoning structure, only in their assumption	Exit check during Part C: observe group work for (1) correct identification of whether the problem is independent or dependent before calculating; (2) correct adjustment of denominator in the dependent (mango) problem — P(second ripe | first ripe) = 6/9, not 7/10; (3) correct formula selection and execution. Students who finish early: 'Write one NEW problem of your own involving either seeds, fruit, or matatu journeys in Kenya that uses the multiplication law. Label whether your events are independent or dependent.' Collect these as

	Dependent Events: Using the maize seed tree, students formalise P(A ∩ B) = P(A) × P(B|A), reading P(B|A) as 'the probability that the second seed germinates GIVEN THAT the first seed already germinated.' Students then solve the original prediction problem from Phase 1 and compare to their earlier guesses. Part C — Group Practice: Groups solve two additional problems: (1) From a crate of 10 mangoes (7 ripe, 3 unripe), two are selected without replacement — find P(both ripe); (2) A matatu on the Nairobi–Thika route and a boda-boda both independently have P(on time) = 4/5 — find P(both arrive on time).	seconds.] Cold-call 2 students — direct them to their tree diagram. 'Your tree tells you: after one G is removed, 2G remain from 4 seeds. So P(B|A) = 2/4. We write: P(A ∩ B) = P(A) × P(B|A) = 3/5 × 2/4 = 6/20 = 3/10.' Box this formula. Say: 'Now check your Phase 1 prediction. Who was closest to 3/10?' [Pause for student reactions.] 'Address the misconception directly: If someone wrote 3/5 × 3/5, ask: 'What did that calculation assume about the second draw? Is that assumption true when seeds are not replaced?' [WAIT TIME: 10 seconds.] Cold-call 3 students. For Part C group practice, circulate and use targeted questions: 'How many ripe mangoes remain after one ripe mango is taken? So what is your new denominator?' After 8 minutes of practice, cold-call one group per problem to present their solution on the board.	— by making the assumption explicit and testable, the teacher converts the error into a learning moment.	formative evidence of transfer.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Conditional Probability Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Multiplication of Fractions by Whole Numbers (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each student writes one sticky note (or index card if sticky notes are unavailable) to add to the class Driving Question Board. The sticky note must complete ONE of the following sentence starters (displayed on the board): (A) 'I can now help Mama Njeri because I know that…' OR (B) 'A new question I have about Mama Njeri's situation is…' Students also review the existing DQB cards from Lessons 1–6, identify ONE card that today's multiplication law directly answers or advances, and draw an arrow from that old card to their new card. The class then does a 3-minute gallery walk of new additions.	Say: 'Take 3 minutes individually. Write your sticky note — use one of the two sentence starters on the board. Be specific: use numbers or formulas if you can.' [Allow 3 minutes of silent writing.] 'Now walk to the DQB. Find a card from a previous lesson that your new knowledge connects to or answers. Stick your card near it and draw a connecting arrow.' [Allow 3 minutes for posting and connecting.] Facilitate a 3-minute gallery walk: 'Walk silently, read three cards that are NOT your own. Place a small tick on any card where you think the student has correctly applied today's multiplication law.' [After gallery walk, select 2–3 notable cards to read aloud.] Say: 'Notice how our	Driving Question Board Cumulative Mapping — the DQB is not merely a display but a living knowledge map. By requiring students to physically connect today's new card to a prior lesson's card with an arrow, this strategy makes the cumulative, building nature of the evidence sequence visible and tangible. The peer-tick gallery walk introduces low-stakes peer assessment, building metacognitive awareness of what correct application of the multiplication law looks like in written form.	Read all sticky notes before the next lesson. Observable indicators of success: (1) Student card uses the phrase 'multiply' or references P(A ∩ B); (2) Student card distinguishes independent from dependent events; (3) Student card connects to Mama Njeri's specific context (e.g., 'I can now calculate the probability that Mama Njeri has a good sales day AND her flour supplier delivers on time'). Flag any card that confuses addition law (from Lesson 6) with multiplication law — address at the start of Lesson 8.

		board is filling up. In Lesson 1 we could only say probability measures likelihood. Today we can calculate the probability that TWO things BOTH happen. How is that more useful for Mama Njeri than what we knew in Lesson 1?' [WAIT TIME: 10 seconds.] Cold-call 2 students.		
Model Building Phase 📹 **VIDEO:** Conditional Probability Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Multiplication of Fractions by Whole Numbers (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students revise their individual Probability Model for Mama Njeri's Stall — a visual model they have been building since Lesson 1. Today they must add a new section labelled 'COMPOUND EVENTS' to their model. In this section, they: (1) Add the multiplication law formulas for both independent and dependent events with a worked example from today's lesson; (2) Revise their model's 'Predictions' section to include at least one compound probability statement about Mama Njeri's situation (e.g., estimating P(good sales day AND quality flour delivery) if these are independent events each with probability 3/5); (3) Write one 'Model Limitation' note — what does the multiplication law still NOT tell Mama Njeri? (e.g., it doesn't tell her how many mandazi to fry, only the probability of joint events). Students share their updated model with a partner for a 2-minute peer review using one prompt: 'Is the distinction between independent and dependent events clear in your partner's model?'	Say: 'Open your probability model — the one you have been building since Lesson 1. Today we add a new section: COMPOUND EVENTS. You have 8 minutes to add your formulas, one worked Mama Njeri example, and your Model Limitation note.' [Circulate and provide targeted prompts: 'What are the two events you are multiplying? Are they independent or dependent? How do you know?' and 'What does your model still NOT explain about Mama Njeri's problem?'] After 8 minutes: 'Share with your partner. Partner's job: ask — Is it clear in this model whether the events are independent or dependent? You have 2 minutes.' [After peer review:] Cold-call 2 students to share their 'Model Limitation' note. Write these on the board under the heading 'What our model still cannot explain.' Say: 'These limitations are exactly what drives our next lessons. What tools might we need to go further?' [WAIT TIME: 12 seconds.] Cold-call 3 students. Leave the 'what our model cannot explain' list on the board as a bridge to Lesson 8.	Cumulative Model Revision with Limitation Articulation — requiring students to add to, rather than replace, their probability model makes conceptual growth literally visible over the lesson sequence. The 'Model Limitation' note is a Boundary Analysis strategy: by articulating what the multiplication law cannot do, students develop a more precise and honest understanding of what it CAN do, and they generate genuine motivation for the next lesson's content. This mirrors how scientists iterate models — adding new evidence while acknowledging remaining gaps.	Collect or photograph 4–5 student models at the end of the lesson. Observable indicators of mastery: (1) Both multiplication law formulas are correctly written and labelled (independent vs. dependent); (2) The worked Mama Njeri example uses correct probability values and correct formula selection; (3) The Model Limitation note demonstrates understanding of what the law computes (joint probability) vs. what it does not (optimal quantity decisions, expected value). Students whose limitation note is vague (e.g., 'it doesn't tell everything') receive a targeted prompt at the start of Lesson 8 to sharpen their thinking.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) MISCONCEPTION CHECK — How many students applied P(A) × P(B) (independence assumption) to the

dependent seed scenario during Phase 1 predictions? Was the misconception-confrontation moment in Phase 3 sufficient to shift their thinking, or do I need to open Lesson 8 with a brief re-engagement on this distinction? (2) TREE DIAGRAM FLUENCY — Were students able to correctly adjust the denominator on second branches without direct telling? If more than one-third of groups kept the denominator at 5, I should design a short warm-up for Lesson 8 that revisits the 'what remains in the bag?' reasoning before extending to three-draw scenarios. (3) DQB QUALITY — Did students' sticky notes use mathematical language (formulas, fractions) or remain vague? If cards were largely qualitative, introduce a 'show your maths' norm for DQB cards from Lesson 8 onward. (4) PACING — The 80-minute lesson has five dense phases. Note which phase required more time than planned; if Phase 2 (tree diagram construction) ran long, consider providing a partially completed second-branch set in Lesson 8 to conserve time for the deeper P(B|A) extension. (5) DIFFERENTIATION OBSERVED — Which students completed the extension task (writing their own problem) in Phase 3? These students are ready for three-event multiplication chains and Bayes' reasoning previews in Lesson 8. Which students still needed scaffolding on single-fraction calculations — prepare a fraction review resource for those students before Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed (via tree diagram construction) that when maize seeds are drawn without replacement, the denominator of the second branch CHANGES to reflect the reduced sample space — the two events are LINKED. In contrast, when flipping a coin twice, the second branch denominators remain unchanged — the events are INDEPENDENT. Students observed that pathway probabilities on a tree diagram are calculated by multiplying the fractions along each branch, and that all pathway probabilities sum to 1.
What did I learn?	Students learned that: (1) When two events are INDEPENDENT, $P(A \cap B) = P(A) \times P(B)$; (2) When two events are DEPENDENT (such as drawing without replacement), $P(A \cap B) = P(A) \times P(B\|A)$, where $P(B\|A)$ reads as 'the probability of B given that A has already occurred'; (3) The key question for determining which formula to use is: 'Does the outcome of the first event change the sample space for the second event?'; (4) For the maize seed scenario (3 germinating, 2 not, from 5), P(both germinate) = $3/5 \times 2/4 = 6/20 = 3/10$.
How does this explain the phenomenon?	Students explained (and can now use to advise Mama Njeri) that: mathematics can calculate the probability of TWO uncertain events BOTH occurring, by multiplying their individual probabilities — but only if you carefully check whether the events affect each other. This is more powerful than single-event probability because Mama Njeri's real business decisions often depend on MULTIPLE things going right at the same time (e.g., good customer turnout AND quality flour supply). The multiplication law gives her — and us — a precise mathematical tool to quantify those compound uncertainties, moving us closer to answering the driving question of how mathematics can support smarter decisions even when no outcome is certain.

LESSON 8 (80 minutes): Probability Tree Diagrams: Mapping Sequential Outcomes	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students construct and interpret probability tree diagrams to systematically calculate the probabilities of sequential (multi-stage) events, multiplying probabilities along branches and adding probabilities across branches.
Knowledge	Students will know that: (1) a tree diagram is a visual tool that maps every possible outcome of a sequential experiment across stages (branches); (2) the probability of a path through the tree is found by multiplying the probabilities along its branches; (3) the probability of an event spread across multiple paths is found by adding the path probabilities; (4) all branch probabilities from a single node must sum to 1; (5) tree diagrams apply to both independent and dependent (non-replacement) sequential events.
Skills	Students will be able to: (1) draw a correctly labelled two-stage and three-stage probability tree diagram; (2) assign correct probabilities to each branch; (3) calculate the probability of a combined outcome by multiplying along branches; (4) calculate the probability of an event that spans multiple branches by adding path probabilities; (5) use tree diagrams to solve real-life sequential probability problems set in Kenyan contexts.
Attitudes	Students appreciate that organising information visually and systematically reduces errors and builds confidence in handling uncertainty — a disposition that supports sound decision-making in everyday Kenyan life, from market trading to agriculture.
Key Inquiry Question	How do we determine the probability of an event? How do we use probability to make decisions in real-life situations?
Purpose in Storyline	In Lessons 1–7 students established the theoretical and experimental foundations of probability and explored combined events using tables and the addition/multiplication rules. In this penultimate evidence-gathering lesson, students add the most powerful visual tool in their probability toolkit: the tree diagram. By re-examining Mama Njeri's mandazi scenario and the matatu/seed germination problems through tree diagrams, students see how sequential uncertainty can be mapped, calculated, and communicated clearly — bringing them one step closer to giving Mama Njeri a fully justified, mathematics-backed recommendation.
Safety Notes	No physical hazards. Ensure pair/group work is inclusive; circulate to check that no student is left passive. If using printed tree diagram templates, ensure students with visual difficulties have enlarged copies.

B. LESSON OVERVIEW
Students open by predicting the probability of a two-day sequence of 'good days' at Mama Njeri's stall using only intuition, then construct a full probability tree diagram for that scenario to test their predictions. They observe how the tree structure forces systematic thinking, practice multiplying along branches and adding across branches, then extend to a three-stage seed-germination problem and a matatu-boarding scenario. By the end of the lesson students can build, label, and calculate from tree diagrams for two- and three-stage experiments — equipping them to tackle any sequential probability question in the driving question context and beyond.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Students are presented with a short story recap: Mama Njeri's	Display the prompt on the board: 'Mama Njeri's records show	Prediction-Before-Instruction (Anticipatory Thinking): By	Observable indicator: Students produce a written numerical

📹 **VIDEO:** [Addition of Sequential Capacities - Word Problems] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	notebook shows that on any given day, the probability of a 'good day' (selling ≥ 80% of her mandazi) is 0.6 and a 'bad day' is 0.4. Students are asked to predict — WITHOUT calculating — the probability that she has (a) two good days in a row, (b) exactly one good day out of two consecutive days, and (c) two bad days in a row. They record individual predictions in their exercise books, then share with a partner. The teacher polls the class by show of hands for each scenario and records the spread of predictions on the board.	P(good day) = 0.6. Without doing any formal calculation, predict the probability of: (a) two good days in a row, (b) exactly one good day in two days, (c) two bad days in a row. Write your best guess as a fraction or decimal.' Allow 3 minutes of silent individual thinking (WAIT TIME: do not accept answers yet). Then say: 'Turn to your partner — compare your predictions and decide together on one answer per scenario.' Allow 2 minutes. Cold-call 3 pairs for scenario (a), 3 different pairs for (b), 2 pairs for (c). Record all predictions in a class tally on the board without affirming or correcting. Say: 'We are going to build a mathematical tool today that will tell us exactly who is right — and more importantly, WHY.'	committing to a numerical prediction before instruction, students activate prior knowledge of multiplication of decimals and combined events from Lessons 5–7. The range of class predictions creates cognitive dissonance — especially for scenario (b), where many students under- or over-count paths — which motivates the need for a systematic visual tool.	prediction for all three scenarios before instruction begins. Listen for students who correctly identify that scenario (b) has more than one way to occur — this reveals readiness for multi-path addition. Flag students who predict P(two good days) > 0.6, as they may not yet apply the multiplication rule to sequential events; these students need targeted scaffolding in Phase 3.
Observe Phase 📹 **VIDEO:** [Addition of Sequential Capacities - Word Problems] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher live-constructs a probability tree diagram for Mama Njeri's two-day sales scenario on the board, narrating every decision aloud. Students copy the diagram step by step into their books. The tree shows: Stage 1 (Day 1) → two branches labelled G (0.6) and B (0.4); Stage 2 (Day 2) → from each Stage 1 branch, two further branches G (0.6) and B (0.4), giving four terminal nodes: GG, GB, BG, BB. The teacher then models multiplying along each branch-path and recording the path probability at the terminal node: P(GG) = 0.6 × 0.6 = 0.36; P(GB) = 0.6 × 0.4 = 0.24; P(BG) = 0.4 × 0.6 = 0.24; P(BB) = 0.4 × 0.4 = 0.16. Students verify the sum: 0.36 + 0.24 + 0.24 + 0.16 = 1.00. Students then observe the teacher model how to answer: 'What is P(exactly one good day)?' by circling the two relevant paths (GB	Say: 'I am going to build a map of every possible two-day outcome for Mama Njeri. Watch carefully — every decision I make has a reason.' Draw the root node and label it 'Day 1'. Ask aloud: 'How many possible outcomes are there for Day 1?' Cold-call 1 student. Draw branches and say: 'I write the probability ON the branch — not at the end.' After completing Stage 1, ask: 'From a Good Day, what can happen on Day 2?' Cold-call 1 student. Complete Stage 2. After writing all four terminal path probabilities ask: 'What should all these terminal probabilities sum to, and why?' WAIT TIME: 12 seconds. Cold-call 2 students. After confirming the sum is 1, say: 'Now watch how I answer the question about exactly one good day.' Circle GB and BG paths in a different colour and say: 'I need ALL the ways exactly one good	Live Modelling with Think-Aloud (Worked Example with Metacognitive Narration): The teacher makes every structural decision visible and verbal, including WHY probabilities are written on branches, WHY multiplication applies along a single path, and WHY addition applies across paths. This directly addresses the most common misconception in tree diagrams — students often add along branches or multiply across paths. Annotating predictions from Phase 1 closes the prediction loop and reinforces the value of the tool.	Observable indicator: Students' copied diagrams have probabilities written ON branches (not at nodes), correct four terminal path probabilities, and a verified sum of 1.00. Circulate during copying — if a student writes path probabilities before branch probabilities, pause the class and address immediately. Listen for students who explain the addition rule using the words 'OR' or 'different ways' — this indicates correct conceptual understanding of the addition step.

	and BG) and adding: 0.24 + 0.24 = 0.48. Students compare this answer to their Phase 1 predictions and annotate their diagrams.	day can happen — these are two separate paths, so I ADD.' Pause and ask: 'Why did I add these two and not multiply?' WAIT TIME: 15 seconds. Cold-call 3 students. Resolve predictions from Phase 1 by pointing to each class prediction on the board.		
Explain Phase 📹 **VIDEO:** Addition of Sequential Capacities - Word Problems Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students work in pairs on two structured tasks that require them to construct and use tree diagrams independently. TASK A (Two-Stage): 'Kamau boards a matatu in Nairobi CBD. The probability that the matatu takes the Thika Road route is 0.7 and the Mombasa Road route is 0.3. If on Thika Road, P(arriving on time) = 0.8; if on Mombasa Road, P(arriving on time) = 0.5. Draw a tree diagram. Find: (i) P(Kamau arrives on time), (ii) P(Kamau does NOT arrive on time).' TASK B (Three-Stage): 'A farmer in Kiambu plants three maize seeds. Each seed independently germinates with probability 0.75. Draw a tree diagram for all three seeds. Find: (i) P(all three germinate), (ii) P(exactly two germinate), (iii) P(at least one germinates).' After pair work (~20 minutes), selected pairs present their diagrams on the board. The class checks each diagram using the rule: 'All branches from one node must sum to 1.' Students then write a two-sentence connection back to Mama Njeri: 'How could Mama Njeri use a tree diagram to plan for two consecutive days?'	Distribute the two tasks (printed or on the board). Say: 'Before you draw a single branch, write down: How many stages? What are the outcomes at each stage? What probability goes on each branch?' Allow 1 minute of individual planning before pairs start drawing. Circulate continuously. When you see a correct diagram, ask the pair: 'How do you know your branches are right? Check your sums.' When you see an error (e.g., adding along branches), crouch to the student's level and ask: 'What does being on THIS branch already tell us? Do we add or multiply to get from start to finish?' — do not give the answer directly. After 20 minutes, cold-call one pair per task to the board. After each presentation, ask the class: 'Agree or disagree — show thumbs.' For every disagreement, say: 'Tell me which branch you would change and why.' For the Mama Njeri connection sentences, cold-call 2 students and ask: 'How does this tree diagram help Mama Njeri make a DECISION — not just calculate a number?'	Collaborative Problem-Solving with Gallery Critique (Pair Construction + Class Verification): Pairs construct diagrams together, reducing individual cognitive load for the structural task. The public board presentation followed by a whole-class agree/disagree check transforms individual work into shared knowledge. The Mama Njeri connection sentence re-anchors the mathematical procedure in the phenomenon, preventing the lesson from becoming purely procedural.	Observable indicators: (1) Task A — students draw two stages with correct conditional probabilities on branches (Thika Road sub-branches: 0.8 on-time, 0.2 not; Mombasa Road sub-branches: 0.5, 0.5); P(on time) = $0.7\times0.8 + 0.3\times0.5 = 0.56 + 0.15 = 0.71$; (2) Task B — students draw three stages with 8 terminal nodes, all summing to 1; P(all germinate) = $0.75^3 \approx 0.422$; P(exactly two) = $3 \times (0.75^2 \times 0.25) \approx 0.422$; P(at least one) = $1 - 0.25^3 \approx 0.984$. Flag any pair whose terminal probabilities do not sum to 1 for immediate reteaching of branch assignment. The Mama Njeri sentence should mention 'two consecutive days' and reference at least one calculated probability.
Driving Question Board (DQB) Creation 📹 **VIDEO:**	Students return to the class Driving Question Board. Each student writes one new sticky note in response to the prompt: 'What can Mama Njeri now calculate	Display the driving question prominently: 'How can we use mathematics to measure and predict the likelihood of uncertain everyday events — like Mama	Driving Question Board Curation (Iterative Knowledge-Building Protocol): The DQB is not a one-time activity — returning to it in every lesson creates a visible	Observable indicator: Each student's sticky note names a specific, calculable probability (e.g., 'P(good day followed by bad day) = 0.6 × 0.4 = 0.24') rather

Addition of Sequential Capacities - Word Problems Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	about two consecutive days that she could NOT calculate before today's lesson?' Students post their notes. The teacher facilitates a 5-minute whole-class read-aloud and clustering of notes into two columns: 'New things we can now CALCULATE' and 'Questions we still have.' Students also update their personal probability vocabulary cards by adding three new terms with definitions and a mandazi-context example: TREE DIAGRAM, MULTIPLY ALONG BRANCHES, ADD ACROSS BRANCHES.	Njeri's mandazi sales — so we can make smarter decisions even when we cannot be certain of the outcome?' Say: 'You have just added a powerful new tool. What specifically can Mama Njeri calculate NOW that she could not calculate in Lesson 7?' Allow 3 minutes silent writing (WAIT TIME: full 3 minutes, no prompting). After posting, read 4–5 notes aloud and ask: 'Does this belong in CALCULATE or STILL WONDERING?' Cold-call students to justify each placement. Point to any sticky note that says something like 'probability of two good days in a row' and say: 'In Lesson 1 we could not do this. Count how many tools we have added since then.' For vocabulary cards, model the first entry: 'Tree diagram — a branching diagram showing all sequential outcomes; e.g., Mama Njeri's two-day good/bad day map.'	record of growing explanatory power. Clustering notes into CALCULATE vs. STILL WONDERING keeps students aware of what remains unresolved before the final synthesis lesson (Lesson 9), creating productive anticipation. Vocabulary card updates build disciplinary language incrementally across the unit.	than a vague statement ('we can predict better'). Vocabulary cards must include a mandazi-context example sentence, not just a definition. Students who write only vague notes need questioning: 'Can you give me a number? What would Mama Njeri multiply or add?'
Model Building Phase 📹 **VIDEO:** Addition of Sequential Capacities - Word Problems Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their cumulative probability model (begun in Lesson 1 and revised in each subsequent lesson). They add a new panel labelled 'Tree Diagram Tool' which includes: (1) a small but complete two-stage tree diagram for Mama Njeri's consecutive-day scenario with all branch probabilities and terminal path probabilities filled in; (2) two annotations — one arrow labelled 'MULTIPLY (same path)' and one arrow labelled 'ADD (different paths)'; (3) a written explanation: 'A tree diagram helps Mama Njeri because…' (minimum 2 sentences connecting the mathematics to her decision about how many mandazi to fry). Students then revisit their model's earlier panels and draw a	Say: 'Open your cumulative models. Today's addition is the tree diagram panel — it does not replace what you built before, it ORGANISES it visually.' Project a model template on the board showing where the new panel sits. Circulate and ask three targeted questions as students work: (1) 'Show me which part of your diagram proves the multiplication rule.' (2) 'Your model should show Mama Njeri making a DECISION — where does the decision appear?' (3) 'If Mama Njeri wants to plan for THREE consecutive days, how many terminal nodes would your tree have? Why?' (Answer: $2^3 = 8$ — push students to generalise: n stages with 2 outcomes each → 2^n terminal	Cumulative Model Revision (Progressive Modelling Protocol): The connector arrow between the multiplication rule panel and the tree diagram panel makes explicit that tree diagrams are not a new concept but a new REPRESENTATION of the multiplication rule. This prevents compartmentalised knowledge and helps students see the unit as a coherent toolkit. The generalisation question (2^n terminal nodes) bridges to combinatorics and prepares students for the synthesis lesson.	Observable indicator: The tree diagram panel contains: correctly labelled branches with probabilities, correct terminal path probabilities that sum to 1, clear MULTIPLY and ADD annotations, and a written explanation that mentions a specific decision Mama Njeri could make (e.g., 'fry more mandazi when P(two consecutive good days) = 0.36'). The connector arrow is present and annotated. Students who cannot answer the 'three consecutive days → 8 nodes' question need to re-examine Stage 2 of the matatu task to see the pattern of branching.

	connector arrow from the 'Multiplication Rule for Independent Events' panel (Lesson 6) to the new Tree Diagram panel, annotating the arrow: 'Tree diagrams make this rule visual and systematic.'	nodes.) In the final 3 minutes, cold-call 2 students to share their written explanation sentences aloud. Ask the class: 'Did they connect the mathematics to a real decision? Thumbs up/down.' Close with: 'In our final lesson, Lesson 9, you will bring every tool — experimental probability, theoretical probability, combined events, AND tree diagrams — together to write Mama Njeri a complete mathematical recommendation. Your model today is your preparation for that.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) PREDICTION ACCURACY — Did students' Phase 1 predictions for scenario (b) (exactly one good day) cluster above or below 0.48? If most predicted below 0.48, students did not yet appreciate that there are TWO paths to 'exactly one good day' — reinforce the ADD-across-paths rule at the start of Lesson 9. (2) BRANCH LABELLING ERRORS — Did any students write probabilities at terminal nodes rather than on branches during Phase 2 copying? If so, the live modelling narration needs to be more explicit about the structural convention. (3) TASK B INDEPENDENCE ASSUMPTION — Did students question whether the three Kiambu maize seeds are truly independent? If so, this is excellent critical thinking — acknowledge it and note that the assumption of independence is stated in the problem. This can seed the discussion of dependent events in future learning. (4) DQB QUALITY — Were sticky notes specific and numerical? If not, the prompt needs to be tightened in Lesson 9 to require a number in every DQB contribution. (5) MODEL CONNECTOR ARROWS — Did students draw the connector from the multiplication rule panel to the tree diagram panel, or did they treat the tree diagram as a standalone new tool? The connection between lessons is the most important conceptual outcome of Lesson 8 — if most students treated it as isolated, open Lesson 9 with a 5-minute model-sharing routine where students explain the arrow to a partner.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed the teacher live-construct a two-stage probability tree diagram for Mama Njeri's consecutive good/bad day scenario, watching how probabilities are written on branches, how terminal path probabilities are found by multiplying along each branch-path, and how the probabilities of multi-path events are found by adding across relevant paths. Students also observed that all terminal path probabilities in the complete tree sum to 1.
What did I learn?	Students learned that a probability tree diagram is a visual tool for systematically mapping all possible outcomes of a sequential (multi-stage) experiment; that the probability of a specific path through the tree equals the product of all branch probabilities along that path; that the probability of an event occurring via more than one path equals the sum of those path probabilities; and that this tool applies to both independent events (e.g., Mama Njeri's consecutive days, Kiambu seed germination) and dependent events with conditional branch probabilities (e.g., Kamau's matatu route and arrival time).
How does this explain the phenomenon?	Students explained how Mama Njeri can use a two-day tree diagram to calculate the probability of two consecutive good days (P = 0.36), exactly one good day (P = 0.48), and two bad days (P = 0.16), and connected this to a concrete decision: because the most likely two-day outcome is exactly one good day (probability 0.48), Mama Njeri should plan her stock levels for a mix of good and

	bad days rather than assuming two good days in a row. Students also explained that tree diagrams make the multiplication rule for combined events visual and systematic, reducing errors in complex multi-stage calculations.

LESSON 9 (80 minutes): Synthesis & Unit Assessment: Helping Mama Njeri	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate and apply all probability concepts learned across the unit in a culminating performance task that advises Mama Njeri on her optimal daily mandazi production strategy.
Knowledge	Students demonstrate understanding that: (1) probability measures likelihood on a scale of 0 to 1; (2) experimental probability is estimated from historical data; (3) the sample space lists all possible outcomes; (4) events can be classified as impossible, unlikely, equally likely, likely, or certain; (5) mutually exclusive and independent events follow distinct addition and multiplication laws; (6) tree diagrams enumerate outcomes of compound events systematically.
Skills	Students are able to: compute experimental probability from a frequency table; construct a sample space; classify events; apply the Addition Law $P(A \cup B) = P(A)+P(B)-P(A \cap B)$ and the Multiplication Law $P(A \cap B) = P(A) \times P(B)$ for independent events; draw and interpret a tree diagram; synthesise probability evidence into a justified real-life recommendation.
Attitudes	Students appreciate the power of mathematical reasoning to support decision-making under uncertainty; they demonstrate perseverance, collaborative responsibility, and honest reflection on the limits of prediction.
Key Inquiry Question	How do we use probability to make decisions in real-life situations?
Purpose in Storyline	This is the capstone lesson. Students draw on every concept introduced in Lessons 1–8 to solve the original anchoring phenomenon: advising Mama Njeri on how many mandazi to fry each morning. By presenting group recommendations with full mathematical justification and reviewing the Driving Question Board, students experience the complete arc — from noticing an unexplained pattern in a market notebook to wielding probability as a precise decision-making tool.
Safety Notes	No physical hazards. Ensure group presentations are conducted in a respectful, psychologically safe environment. Remind students that critique should address mathematical reasoning, not personal effort. If the class uses printed data sheets, handle paper tidily to keep the workspace organised.

B. LESSON OVERVIEW
Lesson 9 is the unit's culminating synthesis lesson. Students revisit Mama Njeri's mandazi sales notebook — the anchoring phenomenon introduced in Lesson 1 — and apply the full toolkit of probability concepts (experimental probability, sample space, event classification, addition law, multiplication law, and tree diagrams) to a structured performance task. Working in collaborative groups, they analyse a provided three-month sales dataset, model key scenarios probabilistically, and produce a written recommendation advising Mama Njeri on her optimal daily batch size. Groups then deliver short presentations, and the class conducts a whole-group Final Explanation that directly answers the driving question. The lesson closes with a ceremonial review of the Driving Question Board: questions that have been answered are marked, and lingering curiosities are honoured as doorways to future learning. A side-by-side comparison of students' Day-1 initial models with their final models makes growth in understanding visible and celebratory.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Before any new instruction, each student independently re-reads the original Mama Njeri phenomenon prompt (displayed on screen or printed on cards) and writes a 3–5 sentence 'Day-9 Prediction Entry' in their science notebook: What is your best mathematical advice to Mama Njeri right now, and which probability concepts support it? Students then do a silent Think-Pair-Share with their group partner, comparing their current thinking to the brief 'Day-1 Prediction Entry' they wrote in Lesson 1. The contrast between naive and now-informed predictions is the emotional hook for the lesson.	Display the anchoring phenomenon text prominently. Say verbatim: 'Before we open any notes or data sheets, I want your best thinking right now. In your notebook, write your advice to Mama Njeri and name at least two probability concepts that back it up. You have four minutes — go.' Set a visible timer. After 4 minutes, say: 'Turn to your partner. Compare what you wrote today with your Day-1 entry. What is the biggest change in your thinking?' Allow 2 minutes of paired talk. Cold-call 3 pairs — choose one strong reasoner, one mid-level, one who struggled earlier in the unit — asking: 'What concept changed your thinking the most?' After each response, WAIT 12 seconds before moving on or probing. Do NOT correct or evaluate yet; affirm with: 'Interesting — hold that thought, we will return to it during our Final Explanation.'	Strategy — Reflective Prediction Journaling: By writing a prediction BEFORE reviewing notes, students activate prior knowledge and create a concrete before-snapshot. Comparing it to their Day-1 entry makes conceptual growth visible, motivating engagement with the performance task that follows.	Observable indicator: Every student has a written Day-9 Prediction Entry that names at least two probability concepts (e.g., 'experimental probability from her notebook data' or 'multiplication law for independent days'). Teacher circulates and scans notebooks during paired talk; flag any student whose entry is blank or only qualitative ('she should fry more') for targeted support during the task.
Observe Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group receives the 'Mama Njeri Sales Dataset' card — a structured summary of 60 days of sales data (drawn from the running dataset used across Lessons 1–8, now presented in complete form). The card includes: (a) a frequency table showing daily sales in bands: Less than 60%, 60–79%, 80–94%, 95–100% of stock sold; (b) a note that Mama Njeri currently fries 50 mandazi each morning; (c) a two-event scenario — 'Event A: sells ≥80% on a given day' and 'Event B: it rains that morning' — with rain frequency data also provided. Students study the table silently for 3 minutes, then complete an 'Observation Log' with three columns: What I Notice (patterns	Distribute data cards face-down. Say: 'Flip your card over. You have three minutes of silent reading — no talking, no writing yet. Just read.' After 3 minutes: 'Now complete your Observation Log. You have four more minutes.' Circulate and listen; take note of groups who immediately jump to computation (redirect: 'Slow down — what are you noticing first?') and groups who seem lost (prompt: 'Look at the frequency column. What fraction of the 60 days fall in each band?'). After logs are completed, cold-call 4 groups — one per quadrant of the room — asking: 'Read me one thing you NOTICED.' Write key observations on the board verbatim, no editing.	Strategy — Notice-Wonder-Connect Log: This three-column structure is a disciplined evidence-gathering routine. 'Notice' anchors students in empirical observation before interpretation; 'Wonder' sustains curiosity and feeds the DQB; 'Connect' activates the conceptual toolkit from prior lessons, making the transition to the performance task feel natural rather than abrupt.	Observable indicator: Each group's Observation Log has at least three distinct observations in the 'Notice' column that reference the data (e.g., 'On 24 out of 60 days she sold ≥80%'), at least one question in 'Wonder', and at least two named concepts in 'Connect'. Teacher checks one log per group during circulation; any group missing the 'Connect' column is asked: 'Which lesson first introduced you to this kind of table?'

	in the data), What I Wonder (new questions), What It Reminds Me Of (connections to lessons 1–8 concepts). Groups share logs aloud before moving to the task.	Then ask the whole class: 'Which of the concepts from our unit appear in this data? Call them out.' Accept and list: experimental probability, sample space, event classification, relative frequency. Say: 'Good. These are your tools. Now let's use them.'		
Explain Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups work through the structured 'Help Mama Njeri' Performance Task — a five-part problem set printed on a task sheet. Part 1 (Experimental Probability): Using the frequency table, calculate P(sells ≥80%), P(sells <60%), and P(60–79%). Part 2 (Sample Space & Event Classification): List the sample space for a randomly chosen day using the four sales bands; classify each event as impossible, unlikely, equally likely, likely, or certain with justification. Part 3 (Addition Law): Given that P(rain) = 0.3 and P(sells ≥80% AND rain) = 0.15, find P(sells ≥80% OR rain). Verify whether selling ≥80% and rain are mutually exclusive and explain. Part 4 (Multiplication Law & Independence): On any two consecutive days, are the sales outcomes independent? If yes, calculate P(sells ≥80% on both Monday AND Tuesday). Part 5 (Tree Diagram & Recommendation): Draw a two-branch tree diagram for Monday and Tuesday outcomes (≥80% vs <80%). Calculate all four branch-end probabilities. Then write a 3–4 sentence recommendation to Mama Njeri stating the optimal batch size with mathematical justification. Groups nominate a 'Presenter', a 'Recorder', a 'Questioner', and a 'Mathematician' role.	Hand out task sheets. Say: 'This is your chance to be the mathematicians Mama Njeri is counting on. Your group has 25 minutes. Every answer must show working — a correct answer with no working earns zero marks today.' Set a 25-minute timer. Circulate in two full loops. During Loop 1 (minutes 1–12): check that every group has started Part 1 and is using the frequency table correctly. For groups dividing wrong: 'How many total days are in the dataset? Use that as your denominator.' During Loop 2 (minutes 13–25): focus on Parts 3–5. Common error prompt for Part 3: 'Are you sure these events are mutually exclusive? Can it rain AND she sells ≥80% on the same day?' For tree diagrams: 'Label each branch with its probability. What must the two probabilities at each fork add up to?' With 5 minutes remaining, say: 'Presenters, read your recommendation aloud to your group and make sure it includes a number — how many mandazi should she fry?' After 25 minutes: 'Pens down. Presenters, you have 3 minutes each. Let's hear your advice.'	Strategy — Scaffolded Multi-Part Problem Solving with Role Differentiation: The five-part structure sequences concepts from foundational (experimental probability) to integrative (tree diagram + recommendation), ensuring all students access entry points regardless of readiness. Role differentiation (Presenter, Recorder, Questioner, Mathematician) distributes cognitive load and ensures accountability. The concrete recommendation requirement forces translation from abstract mathematics to real-world decision-making, closing the phenomenon loop.	Observable indicators by part — Part 1: Probabilities sum to 1 and are expressed as decimals or fractions with correct denominators (out of 60). Part 2: All four events listed; classification uses correct vocabulary with stated reasoning. Part 3: Correct application of $P(A \cup B) = P(A)+P(B)-P(A \cap B)$; student explicitly states events are NOT mutually exclusive because $P(A \cap B) \neq 0$. Part 4: Student justifies independence (consecutive days do not influence each other) before multiplying. Part 5: Tree diagram has two levels, four end branches, all probabilities labelled; recommendation includes a specific batch number (e.g., 'Mama Njeri should fry 45 mandazi') with at least one probability value cited as evidence.

Driving Question Board (DQB) Creation 📹 **VIDEO:** [Mutually Exclusive Events Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups deliver their 3-minute presentations to the class. After all presentations, the teacher facilitates a whole-class 'Final Explanation' discussion — the culmination of the unit's storyline. Students are asked: 'We have now answered the driving question. Let's say it together.' The class reads the driving question aloud in unison. Then, using a structured protocol called 'Claim–Evidence–Reasoning', individual students stand and contribute one sentence at a time to build a shared class answer, with peers adding, qualifying, or extending. The class then turns to the Driving Question Board. Together they move sticky notes from the 'Questions We Have' column to the 'Questions We've Answered' column, reading each aloud and citing the lesson where the answer was found. Unanswered questions are celebrated and labelled 'Doors to Further Learning'. Finally, students do a silent L1 vs L9 Model Comparison: they place their Day-1 initial model beside their most recent model revision and write three sentences describing what changed and why.	Manage presentations: give each group exactly 3 minutes (use a visible timer); after each, invite one 'warm' question from the audience (something they agree with or found clear) and one 'cool' question (something they want clarified). After all presentations, say: 'Now — the moment we have been building toward since Lesson 1. Read the driving question with me.' Lead the unison read. Then: 'Who can give me the first sentence of our class answer using Claim–Evidence–Reasoning? A claim about probability, evidence from Mama Njeri's data, and the reasoning that connects them.' WAIT 15 seconds. Cold-call a student who has not yet presented. Build the class answer sentence by sentence — aim for 4–5 contributions. Write the answer on the board. Then move to the DQB: 'Let's walk through every question we posted across this unit. If we answered it, move it. Call out the lesson number.' For unanswered questions, say warmly: 'This question is still open — it is a door to further learning, and that is something to be proud of.' For the L1 vs L9 model comparison, give 5 minutes of silent reflection. Cold-call 2 students to share one change: 'What did your model in Lesson 1 get wrong or leave out?'	Strategy — Claim–Evidence–Reasoning (CER) Protocol for Final Explanation: CER is a structured argumentation framework that requires students to make a precise claim, cite specific data as evidence, and articulate the logical link between them. Applied here to the driving question, it transforms individual task outputs into a shared, co-constructed class explanation — the highest-order sensemaking activity of the unit. The DQB review provides metacognitive closure by making the journey of inquiry visible and celebrated.	Observable indicators: (1) Every group has delivered a recommendation that includes at least one specific probability value. (2) During the CER build, at least 4 different students contribute a sentence; no sentence is purely procedural ('you multiply the probabilities') — each must reference Mama Njeri's context. (3) The DQB has at least 80% of posted questions moved to 'Answered' with a lesson number cited. (4) L1 vs L9 model comparison entries describe at least one conceptual shift (e.g., 'In Lesson 1 I thought probability was just a guess; now I know it is calculated from outcomes in the sample space').
Model Building Phase 📹 **VIDEO:** [Mutually Exclusive Events Principles] Source: CK-12 🔍 Search ARES	Each student completes their Final Model — a one-page annotated diagram that represents the complete probability story of Mama Njeri's mandazi stall. The model must include: (a) a labelled probability scale (0 to 1) with Mama Njeri's key events positioned on it; (b) a mini sample	Distribute blank Final Model templates (A3 size if available, otherwise A4). Say: 'This is your mathematical portrait of everything you now know about probability. Every section must be labelled and every probability must be calculated — not estimated. You have 10 minutes.' Circulate and	Strategy — Cumulative Annotated Model as Unit Artefact: The Final Model is not a new diagram but the culmination of incremental model revisions made across all nine lessons. Requiring lesson-number annotations makes the learning trajectory explicit and concrete. The Gallery Walk with	Observable indicators: (1) Final Model contains all four required components — probability scale with labelled events, sample space diagram, two-day tree diagram with probabilities summing to 1 at each fork and to 1 across all end branches, and Law of Large Numbers annotation using the

[for similar videos] 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	space diagram for one day's sales; (c) a completed two-day tree diagram with all probabilities labelled; (d) an arrow labelled 'The Law of Large Numbers Effect' pointing from the 'individual day (unpredictable)' side to the 'many days (stable fraction)' side, explaining in one sentence why the proportion of good days stabilises. Students annotate each section with the lesson number where they first learned that concept. The model is submitted as the unit's final artefact. The lesson closes with a 'Probability Wall of Fame' — the teacher posts one exemplary recommendation from each group on the classroom wall, and students do a 3-minute gallery walk to read and place a star sticker next to the recommendation they find most mathematically convincing.	prompt: For the probability scale: 'Where exactly does P(sells ≥80%) sit? Mark it with a cross and label it.' For the tree diagram: 'Check: do your four branch-end probabilities sum to 1?' For the Law of Large Numbers arrow: 'Use the phrase «relative frequency» in your sentence.' With 2 minutes remaining: 'Finish your annotations — which lesson number goes with each section?' Collect models. Then set up the gallery walk: post one recommendation per group. Say: 'You have 3 minutes. Read every group's recommendation. Place your star sticker on the one that most convinces you mathematically — not the prettiest, the most mathematically sound.' After the gallery walk: 'Look at where the stars clustered. What made that recommendation convincing?' Take 2–3 responses. Close with: 'Three months ago, Mama Njeri could not explain why her good-day fraction kept stabilising. Now you can. That is the power of probability — and it is yours.'	star stickers introduces peer evaluation of mathematical argumentation quality, developing critical reasoning skills while generating positive social energy to close the unit.	phrase 'relative frequency'. (2) Every probability value on the model matches the calculations from the performance task (internal consistency check). (3) During the gallery walk, students are observed reading — not just glancing — at peer recommendations before placing their star. (4) Post-walk discussion elicits at least one student response that references a specific mathematical feature (e.g., 'That group used the addition law correctly and showed that rain and high sales can happen together').

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) PHENOMENON CLOSURE — Did students explicitly connect their final recommendations back to the original mystery of why Mama Njeri's good-day fraction stabilises? Could every student articulate the role of the Law of Large Numbers in their own words? (2) PERFORMANCE TASK QUALITY — Which of the five parts caused the most difficulty? Was it the addition law (common error: treating non-mutually-exclusive events as mutually exclusive), the independence justification, or the tree diagram labelling? Plan targeted re-teaching for any concept where more than 40% of groups made errors. (3) DQB COMPLETION — Were there significant questions on the DQB that remained unanswered? If so, note them as entry points for Grade 11 probability or statistics lessons. (4) EQUITY OF VOICE — During the CER Final Explanation, did the same three or four students dominate? Review cold-call records and identify students who contributed to the performance task but were silent during the whole-class discussion — create structured talk opportunities for them in the next unit. (5) MODEL GROWTH — Review L1 vs L9 model comparison entries. Do students describe conceptual shifts (understanding of probability as a mathematical measure) or only procedural additions (I learned the addition formula)? Conceptual shift language is the indicator of deep learning. (6) CELEBRATION — Did the lesson end with a genuine sense of achievement? The gallery walk and closing statement are intentional emotional anchors — if time ran short and these were cut, find a brief way to honour student growth in the next lesson or on the classroom noticeboard.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students examined Mama Njeri's complete 60-day sales frequency table and noticed that roughly 40% of days resulted in sales of ≥80% of stock, while the proportion of very poor days (less than 60% sold) was consistently small — a pattern that held even as individual daily outcomes varied widely.
What did I learn?	Students applied the full probability toolkit: computing experimental probabilities from frequency data, constructing the sample space for a day's sales outcomes, classifying events on the 0-to-1 scale, using the Addition Law $P(A\cup B) = P(A)+P(B)-P(A\cap B)$ for non-mutually-exclusive events, applying the Multiplication Law for independent consecutive days, and building a two-day tree diagram — all synthesised into a justified recommendation for Mama Njeri's optimal daily batch size.
How does this explain the phenomenon?	The fraction of 'good days' stabilises over time because, although each individual day's sales outcome is uncertain (a random experiment), the relative frequency of favourable outcomes converges toward the true probability as the number of trials increases — this is the Law of Large Numbers. Probability does not eliminate uncertainty for any single day, but it gives Mama Njeri a mathematically grounded strategy: by frying a batch size calibrated to her experimental probability of high sales, she minimises expected losses across many days, even though she can never be certain about tomorrow.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.